



Increasing solar UV radiation in Dortmund, Germany: data and trend analyses and comparison to Uccle, Belgium

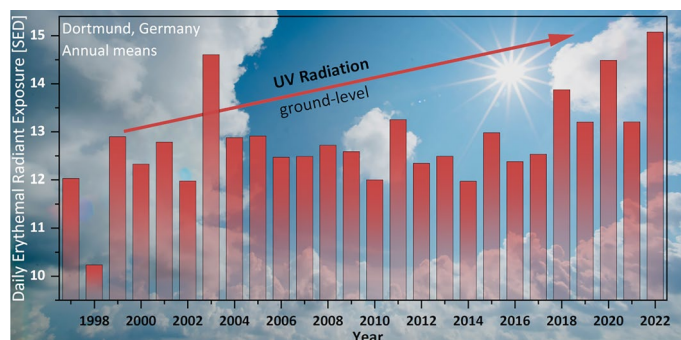
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Abstract

Increasing solar ultraviolet radiation (UVR) can raise human exposure to UVR and adversely affect the environment. Precise measurements of ground-level solar UVR and long-term data series are crucial for evaluating time trends in UVR. This study focuses on spectrally resolved data from a UVR measuring station in Dortmund, Germany (51.5° N, 7.5° E, 130 m a.s.l.). After a strict quality assessment, UV data, such as the daily maximum UV Index (UVI_{max}) and daily erythemal radiant exposure ($H_{er,day}$) values, were analyzed concerning monthly and annual distribution, frequency, occurrence of highest values and their influencing factors. An advanced linear trend model with a flexible covariance matrix was utilized and applied to monthly mean values. Missing values were estimated by a validated imputation method. Findings were compared to those from a station in Uccle, Belgium (50.8° N, 4.3° E, 100 m a.s.l.). Parameters possibly influencing trends in both UVR and global radiation, such as ozone and sunshine duration, were additionally evaluated. The 1997–2022 trend results show a statistically significant increase in monthly mean of $H_{er,day}$ (4.9% p. decade) and UVI_{max} (3.2% p. decade) in Dortmund and $H_{er,day}$ (7.5% p. decade) and UVI_{max} (5.8% p. decade) in Uccle. Total column ozone shows a slight decrease in the summer months. Global radiation increases similarly to the UV data, and sunshine duration in Dortmund increases about twice as much as global radiation, suggesting a strong influence of change in cloud cover. To address health-related consequences effectively, future adaptation and prevention strategies to climate change must consider the observed trends.

Graphical abstract



Keywords Solar UV radiation · Human health · Trend · UV Index · UV exposure · Ozone

Abbreviations

UVR	UltraViolet Radiation	GR	Global Radiation
UVI_{max}	UV Index, daily maximum	GR_{int}	Daily integral of GR
$H_{er,day}$	Daily erythemal radiant exposure	GR_{max}	Daily maximum of GR
SED	Standard Erythemal Dose (= 100 J/m ² erythemal radiant exposure)	TCO	Total Column Ozone
		SunD	Sunshine Duration
		AOD	Aerosol Optical Depth

Extended author information available on the last page of the article

BfS	Bundesamt für Strahlenschutz, German Federal Office for Radiation Protection
DWD	Deutscher Wetterdienst, German Meteorological Service
KMI	Koninklijk Meteorologisch Instituut, Royal Meteorological Institute of Belgium

1 Introduction

The harmful effects of solar ultraviolet radiation (UVR) on public health can lead to various immediate as well as chronic diseases of the skin and eye [1, 2]. The most serious consequence is skin cancer [3–5]; UVR is classified as "completely carcinogenic" (IARC GHS Cat. 1A). In Germany, the incidence of new skin cancer cases increased from about 180,000 to 330,000 per year between 2003 and 2022 [6], with a 65% rise in fatalities, reaching more than 4,400 deaths in 2022 [7]. In Belgium, the incidence of new skin cancer cases between 2004 and 2022 increased from about 4300 to 14,300 per year [8]. Besides its adverse health effects, exposure to solar UVR can also initiate the synthesis of vitamin D. This issue and further effects of sun exposure on human health and on environment are discussed in comprehensive and current overviews by R.E. Neale et al. [9] and S. Madronich et al. [10]. Both the daily maximum UV Index ($UVI_{\max}$) [11] and the daily erythral radiant exposure ($H_{\text{er,day}}$) [12] are important parameters to characterize the harmful effects of ground-level UVR.

The amount of solar UVR reaching a specific location on the Earth's surface depends on numerous parameters [13]: Besides the altitude and the solar elevation, the most important factors are clouds and cloud cover, the total column ozone (TCO) and aerosols or rather the aerosol optical depth (AOD). At places covered by snow, the albedo/surface reflectivity has also to be considered [14–16]. Acute and long-term changes of the influencing parameters lead to changes in solar UV exposure. These changes, caused by ongoing changes in climate, ozone layer and/or anthropogenic aerosols, have the potential to affect humans, life on Earth and the environment with consequences for the health and well-being of humans and the sustainability of the ecosystem [104].

A major challenge to detect changes in solar UV irradiance at the Earth's surface and their causes is the high variability of the influencing parameters as well as the complex interaction of the change processes [17]. Continuous solar UV monitoring activities and networks can be found in many regions in the world, for instance, in Europe [18, 19], Canada [20], the USA [21], Argentina [22], India [23], Thailand [24], Australia [25] or New Zealand [26]. Most UV monitoring stations are equipped with UV broadband radiometers (BRM) [27–29]. UV monitoring with spectroradiometers (SRM) like in the German UV Monitoring network

or the station Uccle, Belgium, is much more sophisticated, but provides significantly more information with spectrally resolved data [30]. Well-maintained and calibrated SRM provide reliable measurements with lower uncertainty [18, 31] and are particularly suitable for long-term trend analysis.

Previous studies analyzing measured data on long-term trends in solar UVR at the Earth's surface have been performed using individual wavelengths in the UVA and UVB range, as well as using $H_{\text{er,day}}$ or $UVI_{\max}$ values. The findings for Europe vary widely, from no significant trend in Finland [32, 33] to a decreasing trend in the UK [34–36] and an increasing trend in Uccle [33, 37], Thessaloniki [33, 36, 38] and Rome in some months [39]. More studies analyzed short- and long-term-trends of UVR in Europe based on shorter time periods [40, 41], looked at earlier periods than our study [1, 42, 43] or for future projection [44], are based on data reconstruction [45–53] or modelled UVR with satellite products [54, 55]. A recent summary of global studies is provided by Bernhard et al. [56]. Due to differences in evaluated time periods and methodologies, only a qualitative comparison is possible in most cases. Different changes of influencing factors, such as surface reflectivity, cloud cover, AOD, and total column ozone, are locally different and relevant. Surface reflectivity and cloud cover are affected by climate change [44, 57]. In recent decades, the impact of ozone on the long-term trend of UVR in Europe is minor compared to changes in cloudiness and aerosols [33, 35, 58]. While ozone may not be a major driver of trends, it still plays a crucial role in Europe. In the short term, ozone can significantly increase solar UVR at the Earth's surface, especially during low-ozone events, leading to higher UV exposure [59–62]. High-quality long-term measurement series of ground-level UVR in Europe are sparse in relation to the large local differences in the previous study results. Furthermore, the current study overview reveals substantial differences in data quality, data processing, handling of data gaps, and trend analysis methodologies, potentially resulting in different outcomes.

Many studies employ linear trend analysis based on calculated monthly anomalies [37, 39]. Other studies, like Hunter et al. [35], used linear and non-linear models to determine a possible trend. Addressing heterogeneity and autocorrelation is often done by modelling deviations from the seasonal pattern rather than the response values themselves (e.g. [37]). However, a violation of the assumption of uncorrelated and homoscedastic error variables leads to underestimated standard errors of trend estimators. The proposed trend model in this article introduces a solution for this issue.

This study analyses a long-term series of spectral UV measurements from Dortmund, Germany and compares it with data from Uccle, Belgium. The focus is on $UVI_{\max}$ and $H_{\text{er,day}}$ values as they are particularly relevant for human exposure. A new statistically approach and a specific

consideration of the handling data gaps is applied and discussed with respect to its impact on the results.

2 Data

2.1 UV monitoring station Dortmund

In Germany, the Federal Office for Radiation Protection (BfS) operates a nationwide network for solar UV monitoring in cooperation with the Federal Environment Agency (UBA), Germany's National Meteorological Service (DWD) and other associated institutions such as the Federal Institute for Occupational Safety and Health (BAuA). BAuA operates the UV monitoring station in Dortmund (51.5° N, 7.5° E, 130 m a.s.l.). Dortmund is located in the Ruhr area, a densely populated region with 10 million inhabitants. The UV monitoring station has been providing spectrally resolved data on solar UV irradiance since August 1996. The station is equipped with a spectroradiometer (Bentham, DTM 300) measuring the horizontal spectral UV irradiance from sunrise to sunset. A detailed description of the uncertainties of the spectral UV measurements performed with such an instrument can be found e.g. in Fountoulakis et al. [63] or Diémoz et al. [64]. For the Dortmund data set, the diffuser temperature was not corrected as it was not recorded. However, due to diffuser heating, the temperature is stabilized above the change point of the diffuser [65]. On-site audits with the world reference QASUME were not performed. However, the 1000W lamp (LDU 100 H), used to check and re-calibrate the working standard in Dortmund, was directly calibrated against the national standard at the National Metrology Institute (PTB) and periodically checked/re-calibrated every few years (expanded uncertainty of the lamp in the certificates < 1.5%). The PTB-traceable calibration of the measurement system with the working standard was conducted at least once a year and was identical over the entire period. The radiometric uncertainty (estimated to be < 2.0% expanded) is approximately randomly set during calibrations.

In the German solar UV monitoring network, different wavelength steps from 290 to 450 nm are used at all stations with double monochromators. Table 1 shows the steps of spectral wavelength. The high resolution in the range of the 393er Fraunhofer lines allows the wavelength accuracy of each spectrum to be monitored. The high resolution at the ozone edge enhances the measurement accuracy in the UVB range. The UV spectra are measured at intervals of 6 min.

In considering daily values (e.g. UVB daily radiant exposure) in this configuration, the total uncertainty of measurements is estimated to be approximately 2.5% (expanded) after data processing (Sect. 2.4). This is significantly lower than a worst-case total uncertainty of a measurement of a

single wavelength under less favorable conditions (high solar zenith angle and low wavelength), which would be less than 6.9%. The main reasons for the difference are that, for daily integrals or daily maximum values, uncertainties at high solar zenith angle carry less weight and the statistical noise becomes negligibly small, as the daily integrals are based on several thousand individual wavelength measurements.

2.2 UV monitoring station Uccle

Uccle is a residential municipality in the South of the Brussels Region and is located about 230 km from Dortmund and 100 km from North Sea shore. The UV monitoring station (50.8° N, 4.3° E, 100 m a.s.l.) is operated by the Royal Meteorological Institute (KMI) of Belgium. It is equipped with two Brewer ozone spectrophotometers (#016 since 1984 and #178 since 2002; [37]). The instruments are integrated in the European network EuBrewNet [66], in the World Ozone and Ultraviolet Radiation Data Centre (WOUDC) and in the Network for the Detection of Atmospheric Composition Change (NDACC, Brewer016). The instruments perform UV scans with constant wavelength steps of 0.5 nm, from 290 to 325 nm (Brewer #016) and from 287.5 to 363 nm (Brewer #178). For wavelengths above 325 nm (363 nm, respectively), the irradiance values are extrapolated using a theoretical spectrum [67, 68] weighted by the intensity at 325 nm (363 nm, respectively). The measurement interval is around 30 min, a frequency typically used in the Brewer network [66]. Thus, the UVI_{max} according to WHO specifications is calculated directly from the daily maximum value. Brewer #016 and Brewer #178 are calibrated each second year against a NIST traceable 1000W DXW lamp. Each month, the stability of the UV spectral observations of the instruments is checked with 50 W UV lamps, i.e., that it is checked that the UV measurements stay within or better than $\pm 5\%$ uncertainty with respect to the last calibration. If the uncertainty becomes larger than $\pm 5\%$, a new UV calibration file is made, taking into account the last UV lamp checks. A total standard uncertainty of $\pm 5\%$ can therefore be assumed for the Uccle UV time series (including correlated and uncorrelated contributions). The instruments were compared to the world traveling reference unit QASUME in

Table 1 Spectral wavelength steps of the double monochromator system in Dortmund

Wavelength Range[nm]	Measurement Step [nm]	Note
290–320	0.5	ozone edge
320–390	5	
392–394	0.075	distinctive Fraunhofer lines
395–450	5	

2004 [69]. For this study, data from Brewer #016 were used when available. Otherwise data from the in-parallel measuring Brewer #178 were used. Thus, after 2002 data gaps only occurred if both instruments were down at the same time.

2.3 Global radiation, sunshine duration and ozone

Additional datasets of global radiation (GR), sunshine duration (SunD) [70] and TCO were investigated to better understand the potential main drivers of trends in solar UV radiation (TCO, clouds, aerosols). In contrast to UVB radiation values, and consequently erythemal radiation values, GR is largely unaffected by current TCO levels but is primarily influenced by factors, such as aerosol optical depth (AOD) and cloudiness. SunD, on the other hand, is mainly influenced by cloudiness alone. For the location Dortmund, a GR and SunD data set was used measured by the DWD at a meteorological station (DWD ID 1117) in the city of Bochum (10 km from the UV monitoring station). The data set includes data of the hourly sum of global radiation in J/cm^2 and data of hourly sunshine duration in minutes. For the location Uccle, a global radiation data set is used detected by the KMI at the same location as the UV data. The measuring interval of the global radiation in Uccle is 10 min.

The TCO values from the Tropospheric Emission Monitoring Internet Service (TEMIS) are used for the parameter ozone. The TEMIS data set provides the local solar noon ozone from a Multi-Sensor Reanalysis MSR-2 (v2.0–2.2) [71, 72]. The data are available for both locations and without gaps over the entire analysis period.

2.4 Data processing

For data processing of the Dortmund data set, various quality checks were performed by BfS to identify wavelength shifts, irradiance deviations, calibration errors and other technical failures and measurement errors: First, all approximately 1 million spectra were checked for wavelength shift compared to the Fraunhofer lines. A limit of ± 0.15 nm was set as the permissible deviation. In addition, various wavelength ratios were checked over the course of the day and year to find wavelength shifts in the spectra outside the Fraunhofer line regions. The calibration data were evaluated to identify irradiance deviations. An imputation method described by Heinzl et al. [73] supports the search for deviations from the irradiance calibration between calibration steps and the retrospective check of the calibrations. In addition, dark current and scan duration parameters were evaluated for anomalies to identify further technical failures and measurement errors.

Together with the calibration data, erroneous spectra were subsequently corrected wherever reliably possible. The UVA irradiance, UVB irradiance and erythemal irradiance were calculated from the remaining spectra. The resulting daily courses for approximately 9000 days (from 1996 to 2022) were screened individually to identify system failures in the course of the day that would lead to errors in the calculation of the UVI_{max} or a daily integral. UVI_{max} is the daily maximum of a 30-min time average value of the erythemal irradiance [11]. For better comparability of the UVI_{max} values, the UVI_{max} is used with decimal places. For UVI_{max} values, the rejected values are mainly on days with missing measurement at midday. To minimize the exclusion of correct data and maximize the exclusion of calculation errors, no fixed interruption time was chosen for exclusion. Each interruption was considered individually based on the prevailing weather conditions, especially clouds. On cloudless days, the UVI_{max} can be reliably calculated if only spectra at the sun's zenith ± 20 min are available. On days with broken clouds, this time period had to be significantly extended.

During data processing, it was found that the measurements in Dortmund started irregularly after sunrise in the first years of the measurement series. Therefore, the daily erythemal radiant exposure ($H_{\text{er,day}}$) was only calculated for the period from 1.5 h after sunrise to 1.5 h before sunset on all days of the time series (the 1.5 h correspond to a sun elevation of around 10° above the horizon). The integral limits ensure that the daily integrals of the individual years are comparable over the entire measurement period. In case of interruptions, it was decided on a case-by-case basis whether the data could be used, e.g. based on the prevailing cloud conditions. In the Uccle data set, the daily UVI_{max} and the $H_{\text{er,day}}$ are rejected when there was an interruption of 2 h or more of the measurements between sunrise and sunset. The values of $H_{\text{er,day}}$ in this study are given in Standard Erythemal Doses (SED) [12]. One SED corresponds to an erythemal radiant exposure of $100 \text{ J}/\text{m}^2$.

A plausible check was performed for global radiation and sunshine duration values of Dortmund, before they were used. In case of erroneous values or interruptions, all values of the day were rejected. The daily maximum of global radiation (GR_{max}) and the daily integral of Global radiation (GR_{int}) and sunshine duration were calculated from the remaining data set. For Dortmund, GR_{int} and sunshine duration values were calculated analogous to the UV data by integration from 1.5 h after sunrise to 1.5 h before sunset. The calculated daily sunshine duration values are given in hours. For Uccle, the 10-min interval of global radiation values allows the calculation of GR_{max} analogous to the UVI_{max} based on a 30-min

average. The GR_{int} was set from sunrise to sunset which corresponds to the calculation of the $H_{er,day}$ in Uccle.

2.5 Handling data gaps

After data processing, data gaps for both UVI_{max} and $H_{er,day}$ occur on around 12% of the days in the Dortmund. In Uccle, UVI_{max} values were missing for around 1% of days in the period from 1997 to 2022 (1991–2022: 4%) and $H_{er,day}$ values for around 3% of days (1991–2022: 11%). To fill the data gaps of UVI_{max} , $H_{er,day}$, and daily integral of UVB irradiance, the validated mixed imputation method [73] is used with TCO and global radiation as predictors. For the imputation in case of UVA irradiance gaps, only global radiation was used as predictor because the influence of ozone on UVA irradiance is negligible. For trend analysis of global radiation and sunshine duration, gaps in the data sets of these values were filled by an average-imputation method [73].

3 Methods

3.1 Descriptive analysis

The descriptive analysis is focused to characterize the local conditions and correlations of the parameters at the station in Dortmund. It describes the underlying values for the trend analysis and gives a comparison of the Dortmund und Uccle data set. Data without imputation of gaps were used in the descriptive analysis. An exception is the calculation of the

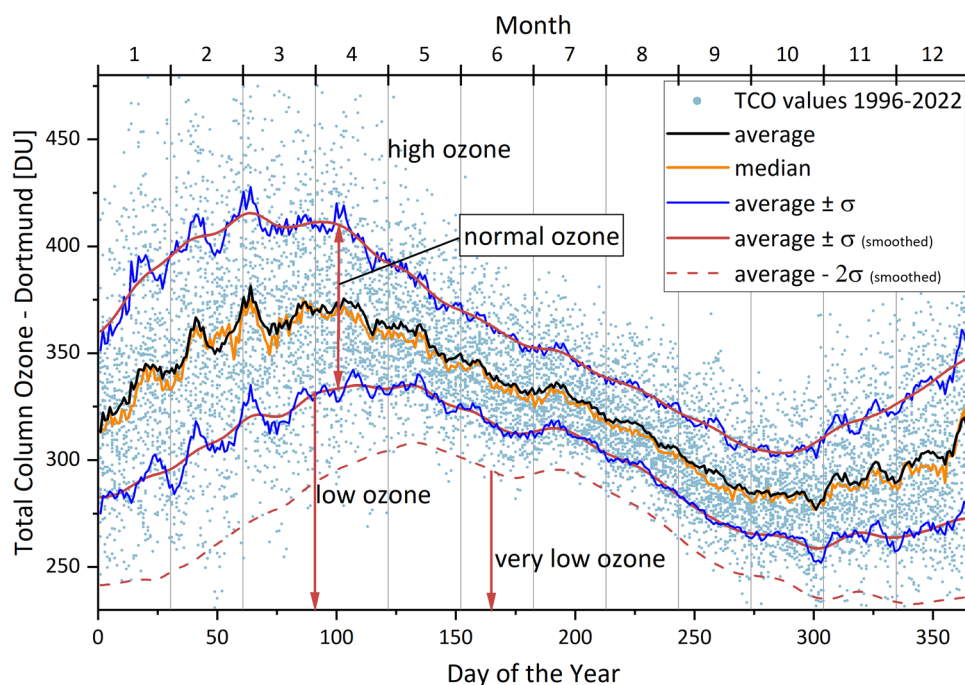
annual mean values in Sect. 4.1.6 and 4.1.7, where inclusion of imputed data is necessary to avoid a bias.

3.1.1 Definition of normal, low and high ozone days

Low ozone has been defined differently in previous studies [46, 59–62, 74–76]. In this study, the annual course of ozone and threshold for low/high ozone is calculated with a simplified approach based on total column ozone values from 1996 to 2022: For each day of the year, the median, the average and the standard deviation σ are calculated from the respective day and all values from the three days before and three days after. The statistics for each day of the year are based on approx. 190 TCO values. The annual curves calculated in this way are shown in Fig. 1 together with the ozone values of all years.

A resulting average $\pm \sigma$ curve is smoothed and serves as a threshold curve for low/high ozone values. Ozone values between smoothed average $\pm \sigma$ are defined as normal, low ozone values as values lower than average $-\sigma$ and high ozone values as values exceeding average $+\sigma$. The resulting low ozone threshold curve corresponds approximately the threshold curve described by Fragkos et al. [74]. For a frequency analysis, the term very low ozone is used to describe low ozone values below average -2σ (Fig. 1, red dashed line).

Fig. 1 Annual course of total column ozone in Dortmund calculated from TEMIS values of 1996–2022 (blue dots): mean (black line), median (orange line), mean $\pm \sigma$ (blue lines) and smoothed mean $\pm \sigma$ (red lines) used as low/high ozone threshold curve



3.1.2 Determination of annual course of potential highest values

The annual courses of potential highest daily values of UVI_{max} and $H_{er,day}$, global radiation and sunshine duration represent a sort of cloudless annual profile, which was calculated on the basis of the long-term measurement dataset. To determine the annual courses of potential highest values, an envelope of all annual courses from 1997 to 2022 is created by identifying the highest measured value for each specific day of the year across all years. The resulting curve is then smoothed to filter out the outlier-days and the days on which there was no cloudless day in all years. To determine the envelope of UVI_{max} and $H_{er,day}$ values, only days with normal ozone are considered to analyze the impact of the normal ozone annual course and low ozone events on the UV data separately. For global radiation (GR_{int} , GR_{max}) and sunshine duration, all days are taken for the calculation. The daily ratio of actual GR_{int} values to a potential highest GR_{int} value serves as indicator for a cloudiness for a specific day. It has to be remarked that limiting the daily integral from 1.5 h after sunrise to 1.5 h before sunset leads to an offset of 3 h to the total day length for sunshine duration and an average annual deviation of 1–2% for $H_{er,day}$ values. This offset also reduces the annual average for GR_{int} by 3.2%, with a standard deviation of just 0.1% for all years of the period.

3.1.3 Correlation analysis over the course of the year

The correlation coefficients of UVI_{max} to total column ozone, GR_{max} and sunshine duration and of $H_{er,day}$ to total column ozone, GR_{int} and sunshine duration were calculated. To analyze the correlations of the variables under all-sky conditions, the correlation coefficients were calculated separately for each day of the year. The resulting correlation coefficients over the course of the year are nearly independent of annual course of solar elevation. The correlation calculation on each day of the year is based on data from the respective day (for all years), plus the data from 5 days prior and 5 days afterward (for all years). For example, to calculate the correlations on September 4th, all available values from August 30th to September 9th from the years between 1996 and 2022 are included. The range of ± 5 days was chosen to suppress sufficiently the annual influence of the solar elevation with short period, but still be able to make a well-founded correlation statement with enough data. Due to the long series of measurement data, each individual calculation of the correlation coefficient is therefore based on 180–291 days. The exact number varies due to gaps in the data sets. For interpretation, a correlation classification by Hinkle et al. [77] was used.

3.2 Trend analysis

We investigated the trend of UV data, global radiation, sunshine duration and total column ozone using a linear model, as it allows us to derive a simple, but quantitative statement of a trend. The trend analysis was performed on the imputed data set for the parameters UVI_{max} , $H_{er,day}$, TCO, global radiation (GR_{max} , GR_{int}), sunshine duration, daily integral of UVA and UVB, and ratio of daily integral UVA to UVB.

The trend analysis is based on monthly values as in other studies (e.g. [37]). In the proposed linear model, called ‘BfS trend model’ in the following, the monthly mean y_t of month t (e.g. of UVI_{max}) increases linearly by β_1 per month, starting at an initial level β_0 . The seasonal fluctuations around the general linear trend are modelled using the effect-coded monthly variables $Jan_t, \dots, Nov_t$ and the corresponding regression coefficients $\beta_2, \dots, \beta_{12}$. The error variable ϵ_t describes the deviation from this pattern for a specific month.

$$y_t = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot Jan_t + \dots + \beta_{12} \cdot Nov_t + \epsilon_t \quad (1)$$

Instead of assuming uncorrelated errors or autocorrelated errors of the order 1 (e.g. [78, 79]), we used a linear trend model that provides a heteroskedasticity and autocorrelation consistent estimation of the covariance matrix implemented in the package ‘sandwich’ for the software R by Zeileis [80] to address heteroscedasticity and autocorrelation in the UV data. The complete deduction of the linear model and covariance matrix can be found in the appendix. Student’s t test for a non-zero trend was used for analysis. Since the t test is robust against deviations from the normal distribution for more than 30 observations, and this condition is satisfied with more than 300 months, a violation of the normal distribution assumption is acceptable.

Results are presented as trend per decade (10 years $\cdot$ 12 month $\cdot \beta_1$) related to the mean value of the analyzed time period to provide stability against year-to-year fluctuations. Additionally, as a reference-independent trend result, we also report the absolute trend per decade.

4 Results

Measured UV data from approximately 8400 days at the Dortmund station, over a period from August 1996 to December 2022, were used for analysis (Fig. 2a, daily UVI_{max} values (left) and $H_{er,day}$ values (right)). Occurring data gaps are unevenly distributed and vary in length. To avoid bias due to these gaps, a data set with imputed data (Fig. 2b) is used for the trend analyses and the calculation of the annual mean values in Sects. 4.1.6 and 4.1.7. Since the incomplete year 1996 could

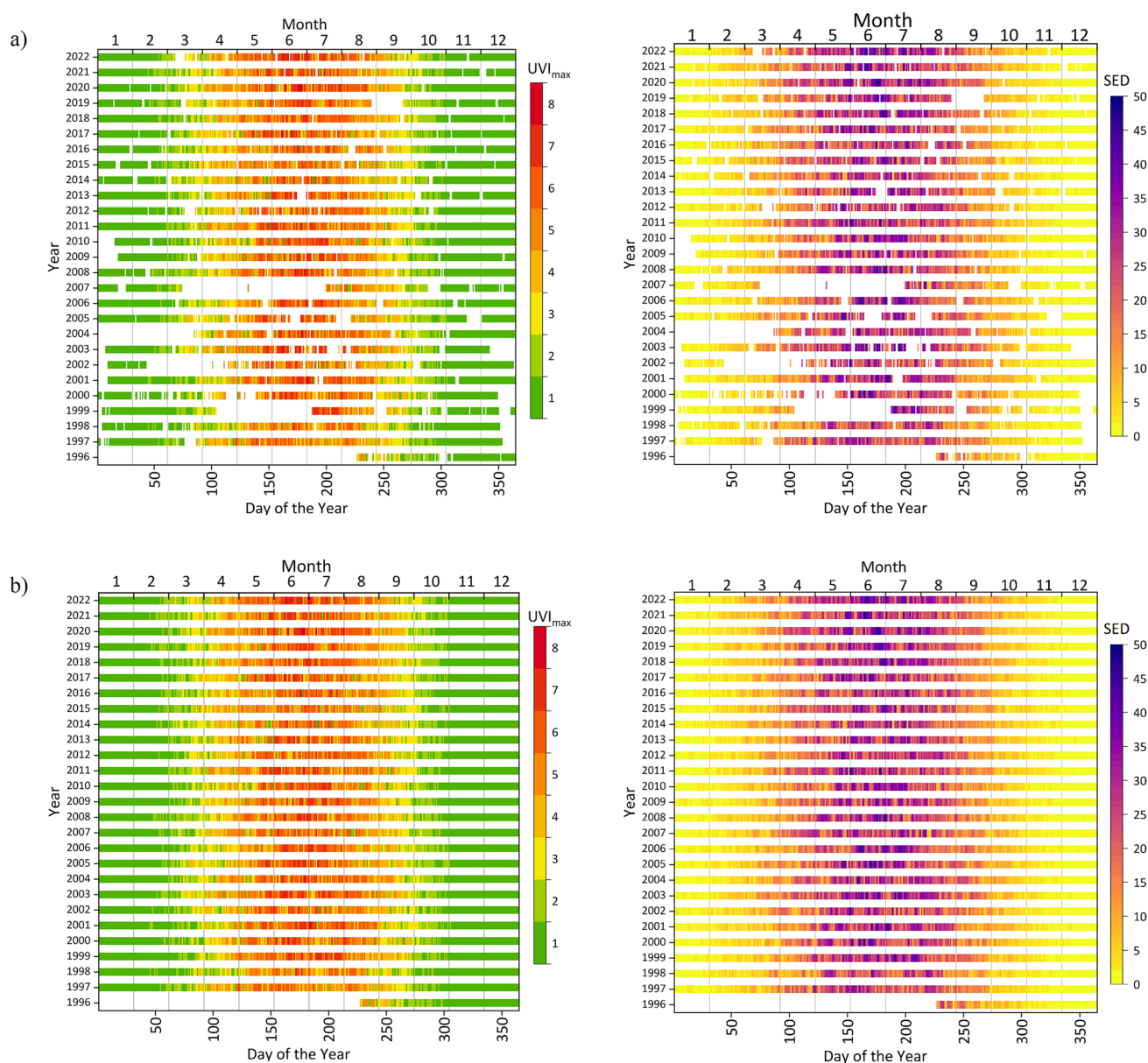


Fig. 2 Heatmap of UVI_{max} values (left) and $H_{er,day}$ values (right) based on measured data from Dortmund, **a** without imputation of missing values and **b** after imputation of missing values. Each row corresponds to a single year, and within each row, the colored lines depict the daily values. The UVI_{max} value colors are chosen according

to the WHO UV Index color scale [6, 11], $H_{er,day}$ values are displayed using a continuous color scale. 97% of the imputed values are derived from available TCO and GR values (model-based imputation), just 3% from statistical means (average-based imputation) [73]

also lead to a bias in the annual values and trend analysis, data from January 1997 onwards are used for these analyses.

4.1 Descriptive analysis

4.1.1 Monthly distribution of UV data and annual frequency of UVI_{max} and $H_{er,day}$

Box plots and density estimates for daily $H_{er,day}$ values (Fig. 3) and UVI_{max} values (Fig. 4) grouped by month

demonstrate that the highest values in Dortmund occur in the months of June and July, strongly correlated with solar elevation. At greater solar elevation, an increased absolute dispersion can be observed. The $H_{\text{er,day}}$ and UVI_{max} datasets from Uccle have a similar monthly distribution. Consequently, for comparative purposes between the

Dortmund and Uccle datasets, only the monthly mean for the Uccle data is illustrated as a grey curve in Fig. 3 and Fig. 4. For both locations, on average 83% of the annual cumulative $H_{\text{er,day}}$ is detected from April to September. UVI_{max} values ≥ 3 (2.5) occur mainly (94%) between April and September.

Fig. 3 Daily erythemal radiant exposure $H_{\text{er,day}}$ from Dortmund 1996–2022 grouped in months, presented as box plots (box: interquartile range, black lines: median, crosses: mean, Whisker: minimum and maximum data distribution within 1.5 times the interquartile range) with density estimates; Grey curve: monthly mean of $H_{\text{er,day}}$ values from Uccle 1996–2022. The statistics are based on 600–800 values for each month

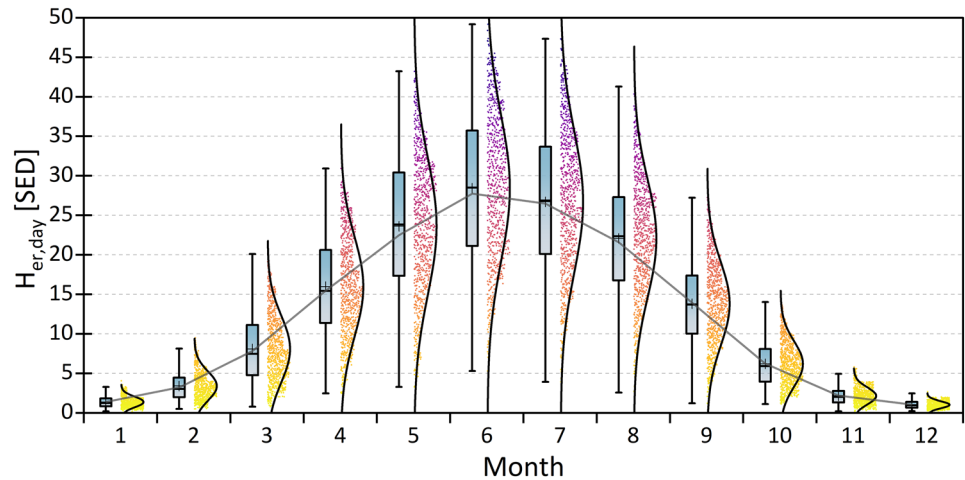


Fig. 4 UVI_{max} values from Dortmund 1996–2022 grouped in months, presented as box plots (box: interquartile range, black lines: median, crosses: mean, Whisker: minimum and maximum data distribution within 1.5 times the interquartile range) with density estimates; grey curve: monthly median of UVI_{max} values from Uccle 1991–2022 for comparison

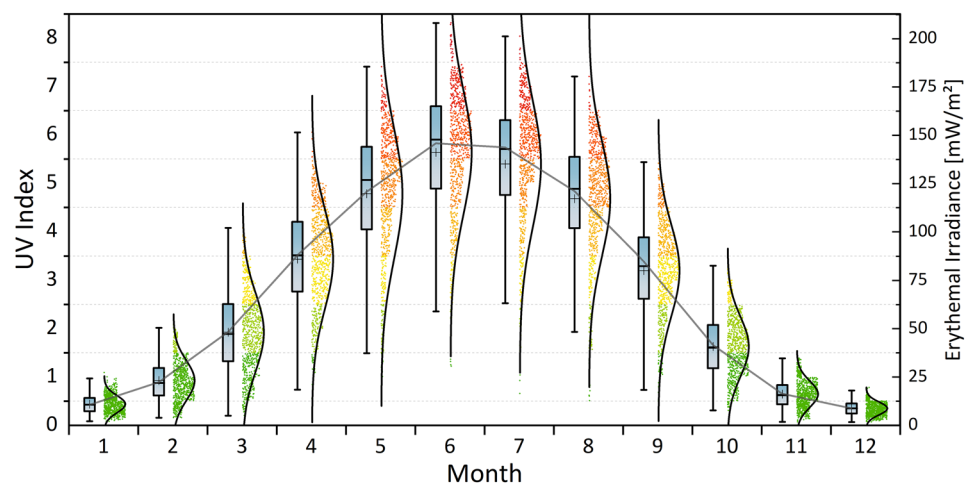


Fig. 5 UVA daily radiant exposure values (315–400 nm) from Dortmund 1996–2022 grouped in months, presented as box plots (box: interquartile range, black lines: median, crosses: mean, Whisker: minimum and maximum data distribution within 1.5 times the interquartile range) with density estimates

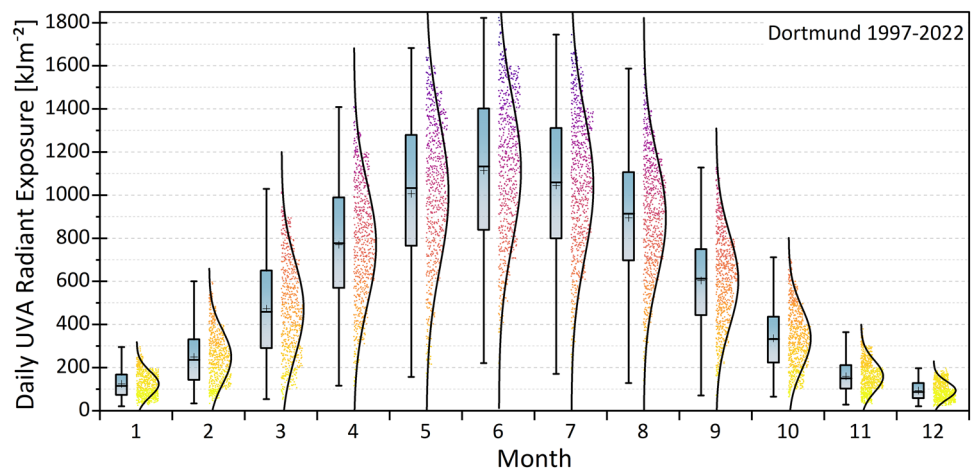


Fig. 6 UVB daily radiant exposure values (290–315 nm) from Dortmund 1996–2022 grouped in months, presented as box plots (box: interquartile range, black lines: median, crosses: mean, Whisker: minimum and maximum data distribution within 1.5 times the interquartile range) with density estimates

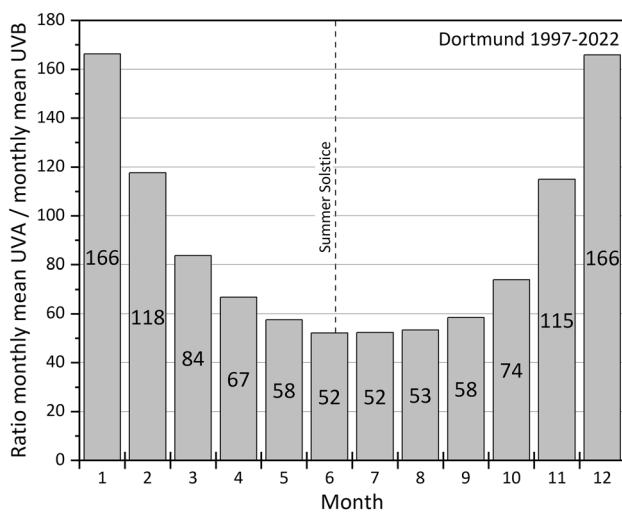
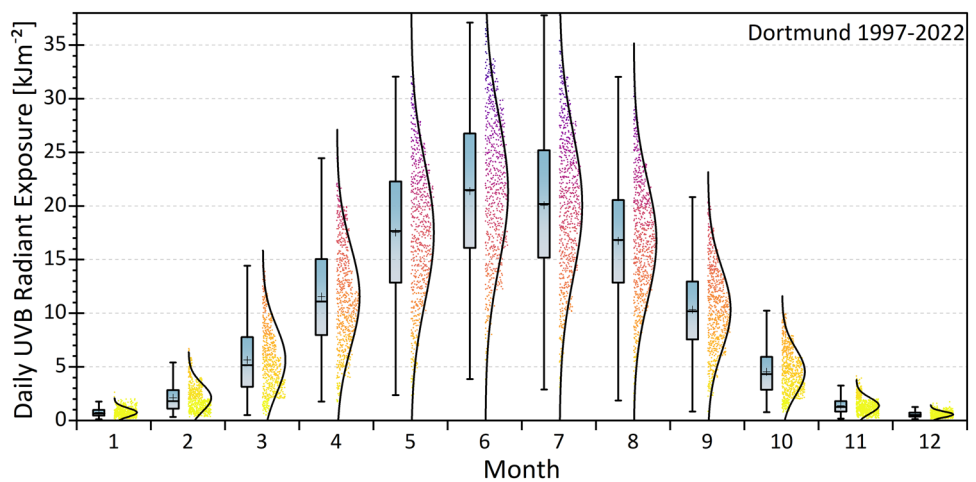
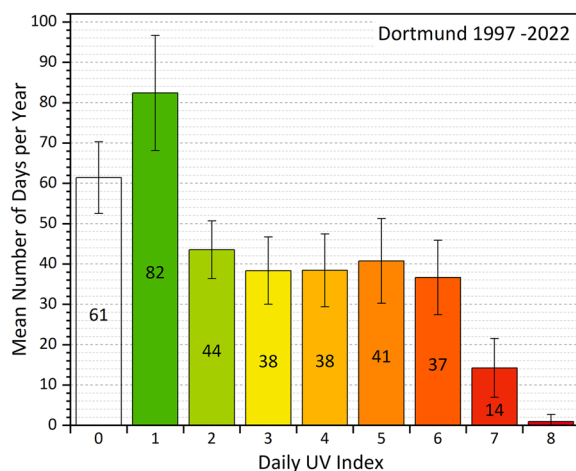


Fig. 7 Ratio of monthly mean daily radiant exposure UVA (Fig. 5) to monthly mean daily radiant exposure UVB (Fig. 6)



Corresponding plots for daily integral of UVA (Fig. 5) and UVB (Fig. 6) show a similar behavior in the annual course compared to $H_{\text{er,day}}$. The Ratio of UVA to UVB (Fig. 7) highlights the differences between these values.

The proportion of UVB radiation in UVR is three times higher in summer than in winter and remains relatively constant from May to September. The annual course is asymmetrical to the summer solstice and shifts toward later months (Fig. 7).

From the annual frequency of daily UVI_{max} values (Fig. 8 left), it can be seen that approximately half of the days in Dortmund have a UVI_{max} value equal to or less than 2. Days with UVI_{max} values of 3, 4, 5, or 6 occur with roughly equal frequency. The number of days with a UVI_{max} 7 is significantly lower, and days with a UVI_{max} of 8 are rare. The frequency analyses of $H_{\text{er,day}}$ shows that approximately one-third of the days fall into the lowest range of 0–5 SED (Fig. 8 right), which corresponds to the number of days with

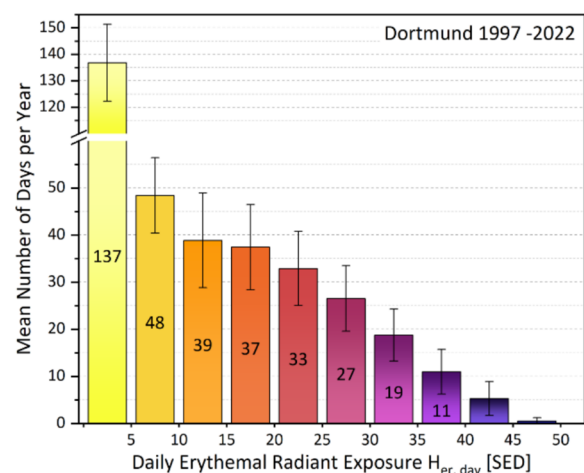


Fig. 8 Mean annual daily UVI_{max} (left) and $H_{\text{er,day}}$ (right) frequency of Dortmund with error bars representing the annual minimum and maximum of occurred number of days from 1997 to 2022

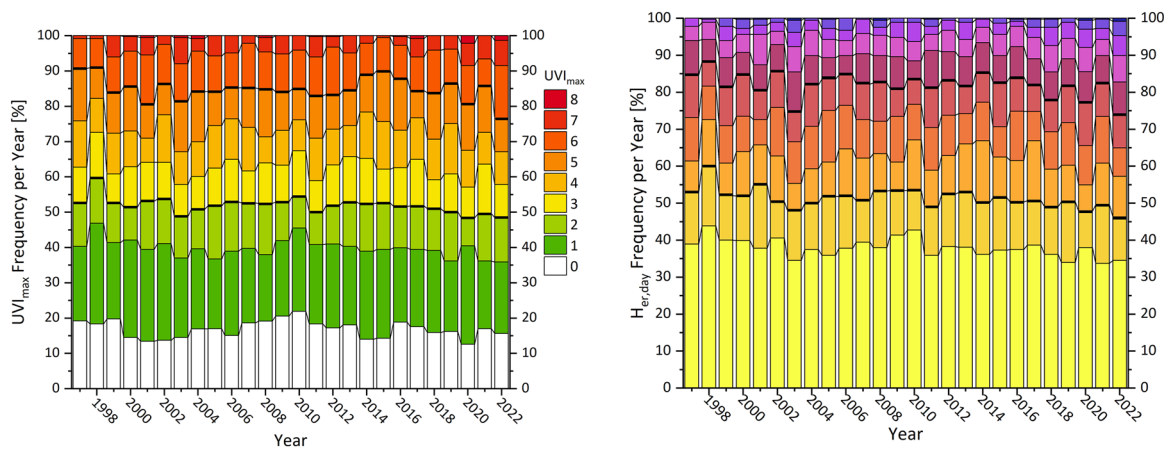


Fig. 9 Time series of the annual UVI_{max} (left) and $H_{er,day}$ (right) frequencies of Dortmund from 1997 to 2022 calculated with imputed data set

UVI_{max} value 0 and 1 (Fig. 8 left). More detailed yearly-resolved frequencies of UVI_{max} and $H_{er,day}$ are illustrated in the Fig. 9. Despite year-to-year variation, a slight tendency of increasing number of days with higher UVI_{max} and $H_{er,day}$ is visible.

4.1.2 Annual course of potential highest values

The annual courses of potential highest $H_{er,day}$ values under normal ozone condition in Dortmund are given in Fig. 10. For comparison, the annual courses of the potential highest daily sunshine duration and the potential highest daily global radiation (GR_{int}) are included in the diagram. It has to be remarked that all daily values are integrals from 1.5 h after

sunrise to 1.5 h before sunset. The course sunshine duration and global radiation are correlated with the solar elevation resulting in a symmetrical curve to the summer solstice. The course of the potential highest $H_{er,day}$ is additionally shifted towards later months. The annual course of median of ozone shows an increase in ozone values between October and March, and a decrease between April and September (Fig. 10, blue line). The ratio of the actual global duration on a day in relation to the maximum potential global duration on this day is used as a parameter for cloudiness in further analyses.

The corresponding plot for the annual course of the potential highest values of daily UVI_{max} , sunshine duration, global radiation (GR_{max}) and median of total column

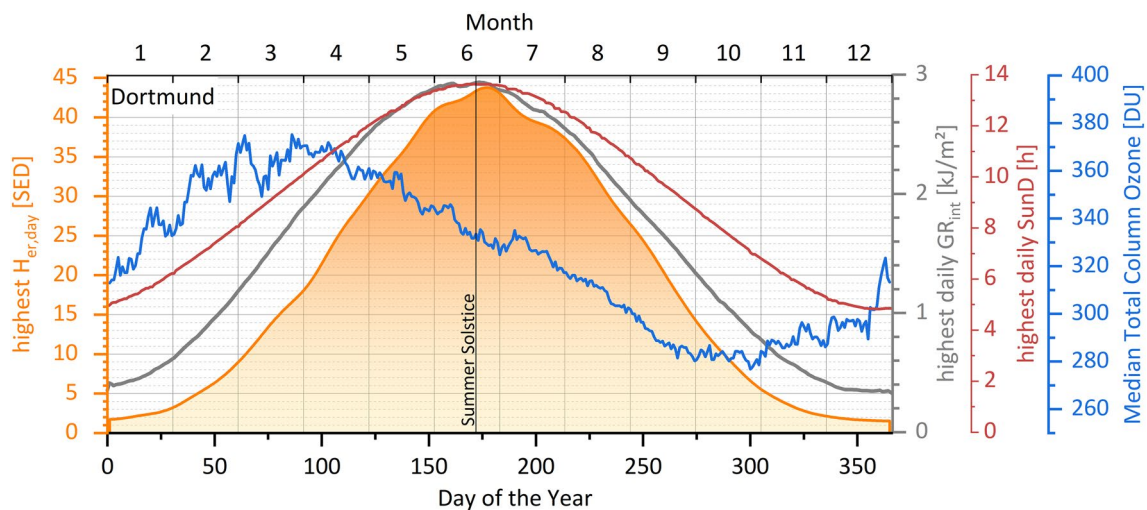


Fig. 10 Annual courses of potential highest daily values at normal ozone conditions for $H_{er,day}$ (orange line, left scale), the sunshine duration (SunD) (red line, third right scale), and the daily global radiation

(GR_{int}) (grey line, first right scale) and the annual course of median of ozone according 3.1.1 (dark blue line, second right scale). Summer solstice is marked by a vertical line

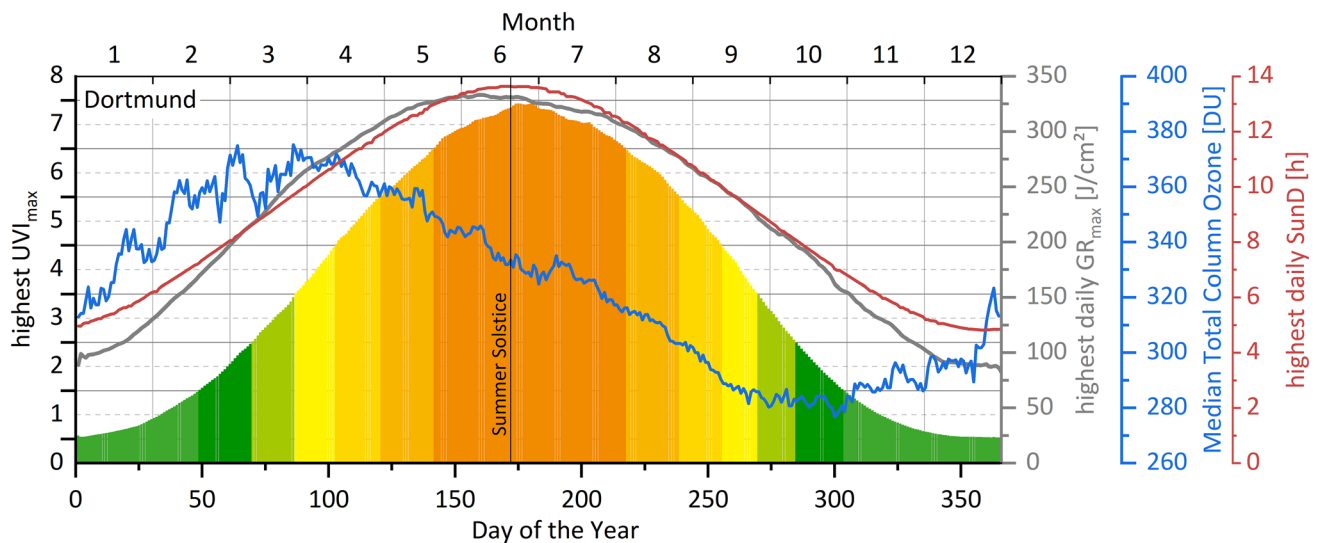


Fig. 11 Annual courses of potential highest daily values at normal ozone conditions for UVI_{max} (colored bars, left scale), sunshine duration (SunD, red line, third right scale), daily global radiation (GR_{max} ,

grey line, first right scale) and the annual course of median of total column ozone according 3.1.1 (blue line, second right scale)

ozone is shown in the Fig. 11. The annual course of GR_{max} values is approximately symmetrical to the summer solstice, whereas the UVI_{max} curve is shifted towards later months. Additionally, a slight attenuation of GR_{max} values from June to September and a slight increase in the preceding months can also be observed in the Uccle data and does not occur in the GR_{int} curve (Fig. 10). The causes of these variances were not investigated in this study.

4.1.3 Correlation analyses of $H_{er,day}$ und UVI_{max} with other variables

Figure 12 visualizes the correlation results of $H_{er,day}$ and UVI_{max} values, respectively, with other variables as the course of the year. $H_{er,day}$ (Fig. 12, left) and global radiation (GR_{int} , blue dots) show a constant very high correlation in the summer months, which is also still high to a very high in the winter months. The correlation of $H_{er,day}$ with the sunshine duration (red dots) is lower but still high. The expected

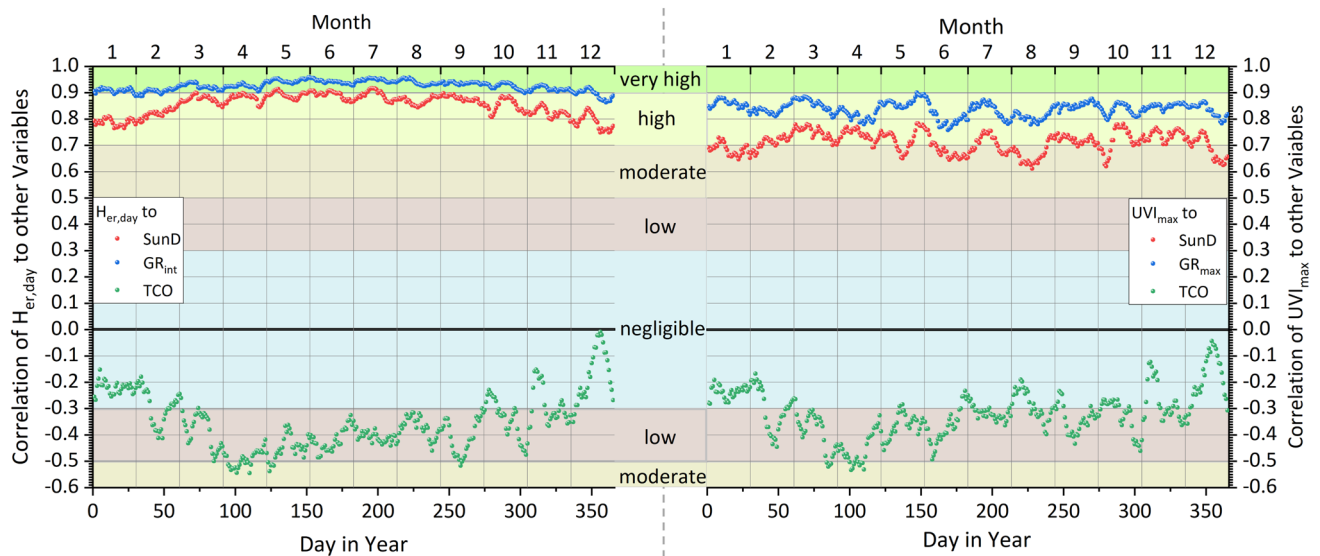


Fig. 12 Correlation coefficients of $H_{er,day}$ (left) and UVI_{max} (right) values with sunshine duration (SunD), global radiation and total column ozone as a course of a year based on data set from the location Dortmund in the period 1996 to 2022 at all-sky-conditions

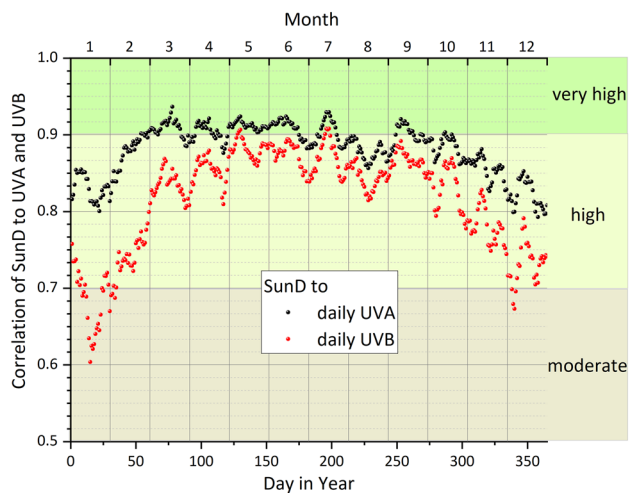


Fig. 13 Correlation coefficients of sunshine duration (SunD) with daily UVA and UVB radiant exposure as a course of a year based on data set from the location Dortmund in the period 1996–2022 at all-sky-conditions

differences arise because SunD only indicates whether a cloud is between the measuring device and the sun (binary response for direct solar radiation). GR, in contrast, can also indicate how much solar radiation passes through a cloud and how much diffuse solar radiation reaching the measuring device. Consequently, the correlation between GR and $H_{\text{er,day}}$ is stronger. Variability in cloud cover, compounded by shorter day lengths in winter, can lead to differences in the correlation coefficient of SunD to UV values throughout the

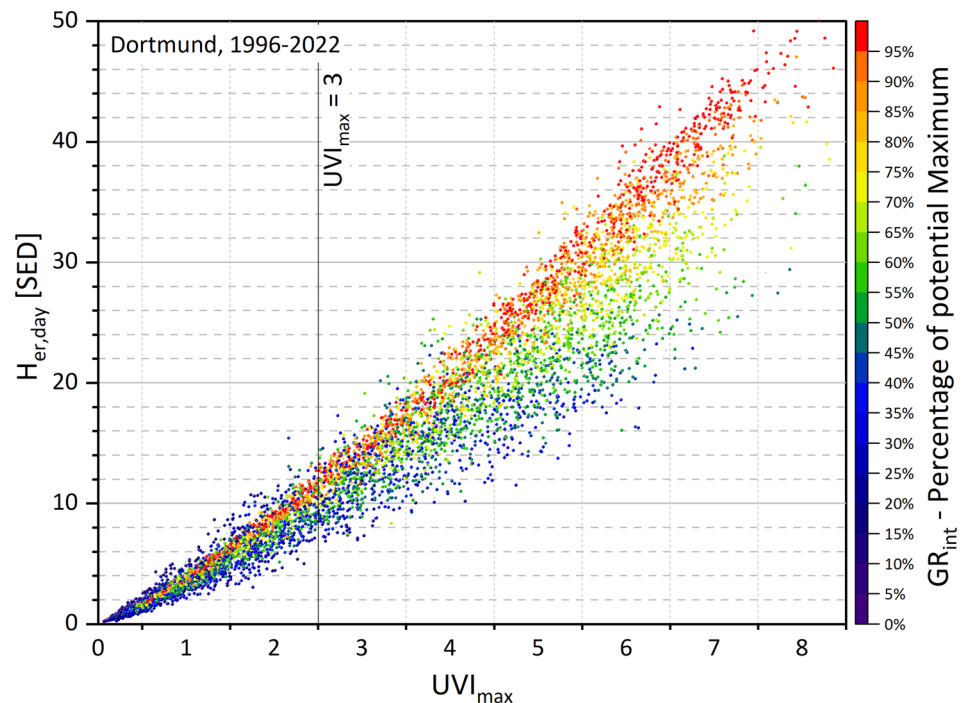
year. UVI_{max} (Fig. 12, right) shows a constant high correlation with the daily maximum values of the global radiation (GR_{max} , blue dots) and a lower but still moderate to high correlation with the daily sunshine duration (red dots). Under the considered all-sky conditions, the correlation of $H_{\text{er,day}}$ and UVI_{max} values with TCO is low and even negligible in the winter months. Correlation of sunshine duration to daily UVA and UVB radiant exposure is high / very high on most days between March and September (Fig. 13). The correlation of sunshine duration to UVB is slightly lower than to UVA. If the correlation coefficients of $H_{\text{er,day}}$ to sunshine duration from Fig. 12 were plotted in Fig. 13, the values would, as expected, fall between those of UVA and UVB.

When plotting all $H_{\text{er,day}}$ and UVI_{max} values from Dortmund against each other in a scatterplot and marking each value with a color according to the prevailing cloudiness on that day (Fig. 14), it can be observed that days with low or no cloudiness (red dots) are at the upper boundary of the pattern and show an approximate linear relationship between $H_{\text{er,day}}$ and UVI_{max} . It also shows, that on days with $\text{UVI}_{\text{max}} < 3$, $H_{\text{er,day}}$ values can exceed 10 SED on some days.

4.1.4 Condition for highest values of UVI_{max} and $H_{\text{er,day}}$

Examining the main influencing factors (solar elevation, total column ozone and global radiation) for the days of the 200 highest $H_{\text{er,day}}$ values in the entire measurement series from Dortmund (approximately 9000 days), it can be observed that on these days high solar elevation, low ozone levels, and low cloud cover coincide:

Fig. 14 Scatterplot of $H_{\text{er,day}}$ and UVI_{max} values measured at Dortmund (1996–2022) with colored dots indicating cloudiness by the ratio of the actual global radiation (GR_{int}) to the calculated daily potential highest global radiation (Fig. 10). Higher values of the ratio (red dots) correspond to cloudless conditions, while lower values (blue dots) indicate overcast skies. The vertical line in the plot marks the range of $\text{UVI}_{\text{max}} \geq 3$



- On 93% of these days, there was below-average ozone (ozone < annual mean total column ozone level).
- On 42% of these days, there was even low ozone (ozone < annual mean- σ total column ozone level).
- On 90% of these days, the ratio of actual global radiation (GR_{int}) to potential highest global radiation is higher than 87%.
- 99% of these days occurred between 30 days before and 41 days after the summer solstice.

For the 200 highest UVI_{max} values, the conditions regarding ozone and the distance of the days to the summer solstice are nearly identical to the highest $H_{er,day}$ days. Additionally, the ratio of actual global radiation (GR_{max}) to potential highest global radiation is higher than 85% on 90% of days with the highest UVI_{max} .

The distribution of days with the highest 200 UV values across the first half (1997–2009) and second half (2010–2022) of the data series is as follows: Of the days with the highest UVI_{max} values, 98 occurred in the first half and 102 in the second half. Regarding the days with the highest $H_{er,day}$ values, 87 occurred in the first half and 113 in the second half.

4.1.5 Low ozone events

Low ozone events can significantly increase ground-level solar UVR, especially in absence of clouds. Figure 15 presents UVI_{max} values for selected periods in Dortmund, during which these low ozone events occurred. At similar solar elevation, the UVI_{max} values increase with increasing negative deviation from the ozone mean value. The exemplary low ozone events shown in Fig. 15 ranged from 10 to 19% in the months from April to July. For the low ozone event in February 2023, an increase of 53% is detected. A separate plot for Uccle has been omitted because, within the limits of measurement accuracy, the increases in UVI_{max} are identical at low ozone events with comparable cloudiness.

Observing the frequency of days with low ozone (mean- σ) in the radiation protection-relevant summer months (Apr–Sept) with the condition of little to no clouds at mid-day (global radiation $GR_{max} > 90\%$) in Dortmund, it can be seen that in the second half of the period (2010–2022) there occur about 8% more days (165 days) than in the first period (1997–2009: 153 days). If we focus only on the days with very low ozone (mean-2 σ), their frequency increases from 6 days (1997–2009) to 16 days (2010–2022).

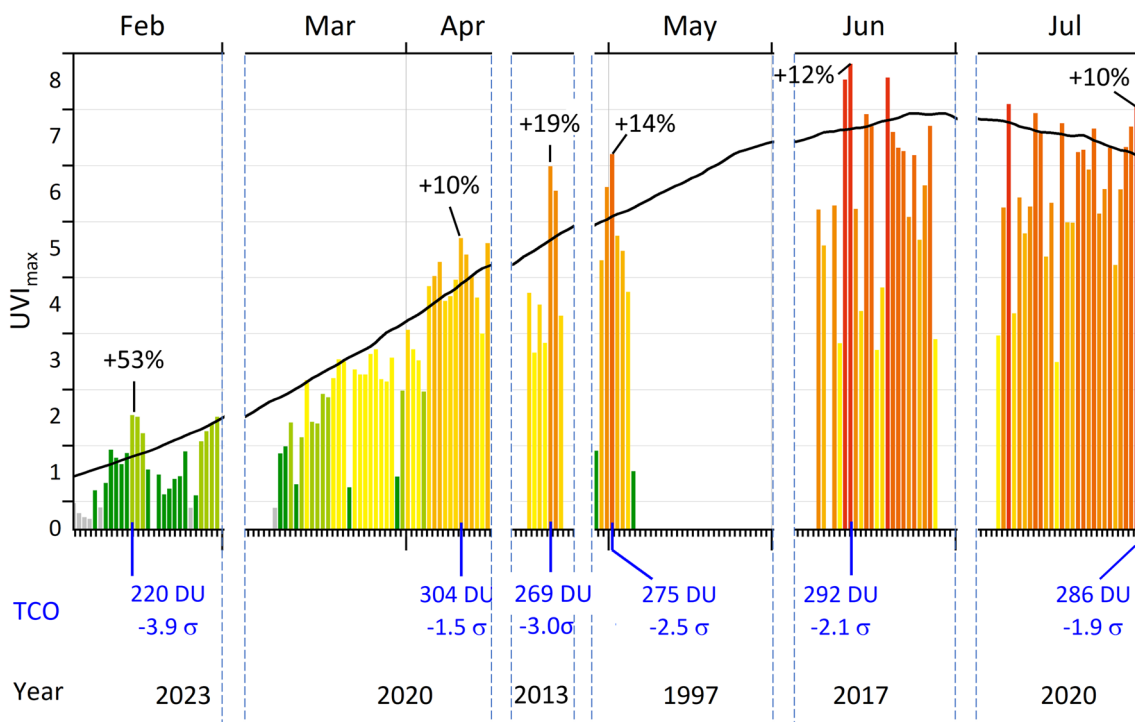


Fig. 15 Daily UVI_{max} values (colored bars) for exemplary low ozone events separated by dashed lines; percentage increase of the highest UVI_{max} value within each period relative to the potential highest UVI_{max} value (Fig. 11), and respective total ozone column value

of these days with the number of standard deviations below mean according to 3.1.1 (blue), and the year of the occurring low ozone event

4.1.6 Annual mean of $H_{\text{er,day}}$ compared to GR and TCO

Ground-level solar UV values and global radiation, as well as atmospheric ozone levels, exhibit year-to-year fluctuations. To address the question of whether years with high global radiation and/or low ozone levels correspond to high annual UV radiation, the annual values of $H_{\text{er,day}}$ (red bars), global radiation (GR_{int} , black line) and TCO (blue line, inverted scale) are compared (Fig. 16). For the ozone data, the annual median of daily summer ozone values (Apr–Sept) is chosen since the majority of annual UVR occurs in the summer and thus the ozone values in summer have a greater influence on annual UVR.

The comparison shows that in years with exceptionally low ozone values, the annual $H_{\text{er,day}}$ increases significantly. This can be seen comparing the years 2018 and 2020: In 2018, the annual mean of global radiation exceeds that of 2020, implying higher $H_{\text{er,day}}$ values in 2018. Due to the very low ozone levels in 2020, the $H_{\text{er,day}}$ values for that year were additionally elevated and exceed the values of 2018. Even in years with above-average ozone levels, the ozone influence is discernible. For instance, in the year 2010, the annual value of $H_{\text{er,day}}$ decreases, contrary to expectations based solely on global radiation.

4.1.7 Station comparison

After comparing the annual mean values of $H_{\text{er,day}}$ from Dortmund to the influence factors global radiation and ozone in chapter 4.1.6, the next step involves comparing data from Dortmund and Uccle. This aims to identify differences in annual UV radiation values and investigate how differences can be traced back to differences in the influencing factors.

For both stations, annual values of $H_{\text{er,day}}$, UVI_{max} , summer TCO (Apr–Sept), and global radiation (GR_{int}) were compared (Fig. 17).

It can be observed that all annual values from both locations are generally in the same level. The differences occurring in $H_{\text{er,day}}$ in a few years are also visible in the global radiation data in the same years. Black circles highlight periods where global radiation and $H_{\text{er,day}}$ in Dortmund exceeded Uccle, green circle indicates an exemplary period during which global radiation and $H_{\text{er,day}}$ in Uccle exceeded Dortmund. The annual ozone data from Dortmund and Uccle differ hardly.

To illustrate the impact of the different measurement intervals (6 min in Dortmund, approx. 30 min in Uccle) on UVI_{max} data, the dashed blue line represents the annual mean values from Dortmund based on a 30-min analysis. On average, this adjustment results in a 1.7% increase in the annual UVI_{max} values from Dortmund. To illustrate the difference between annual summer TCO values and annual TCO values (all month), the latter are additionally plotted for Dortmund with a dashed blue line.

4.2 Trend analyses

The trend analysis is based on monthly mean values of imputed data. Correcting the data gaps with imputed data (Fig. 2) results in remarkable changes for some of the monthly means (Figs. 18 and 19, grey and numbered circles). The linear trend is visualized with a dashed red line and shows an increase over the 26 years. The seasonal course around the trend with the highest value in June and the lowest value in December is shown by a solid red line. Years with low (e.g. 1998, 2007, 2012) and high (e.g. 2003,

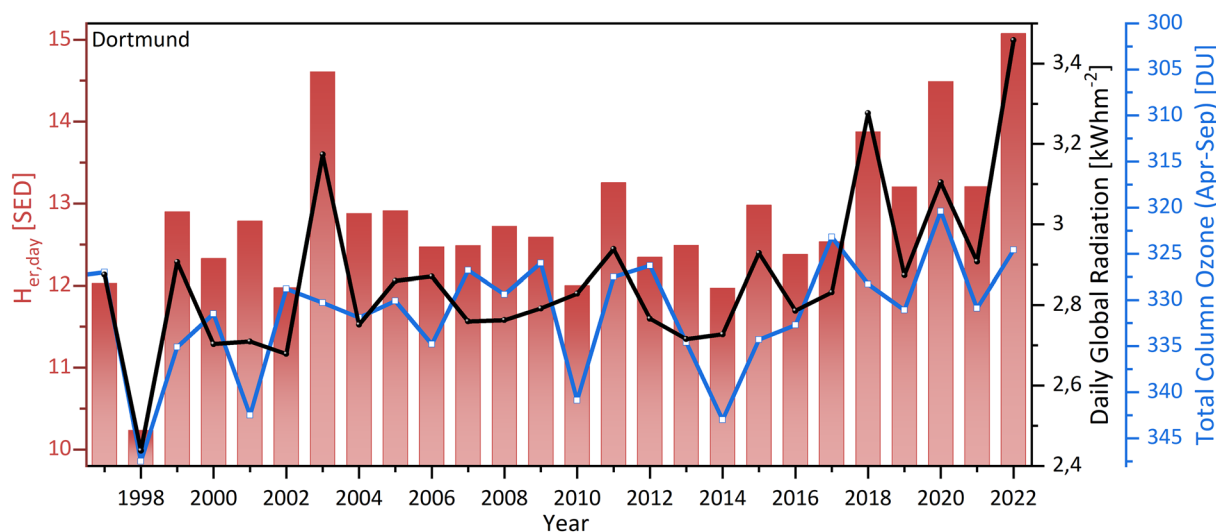
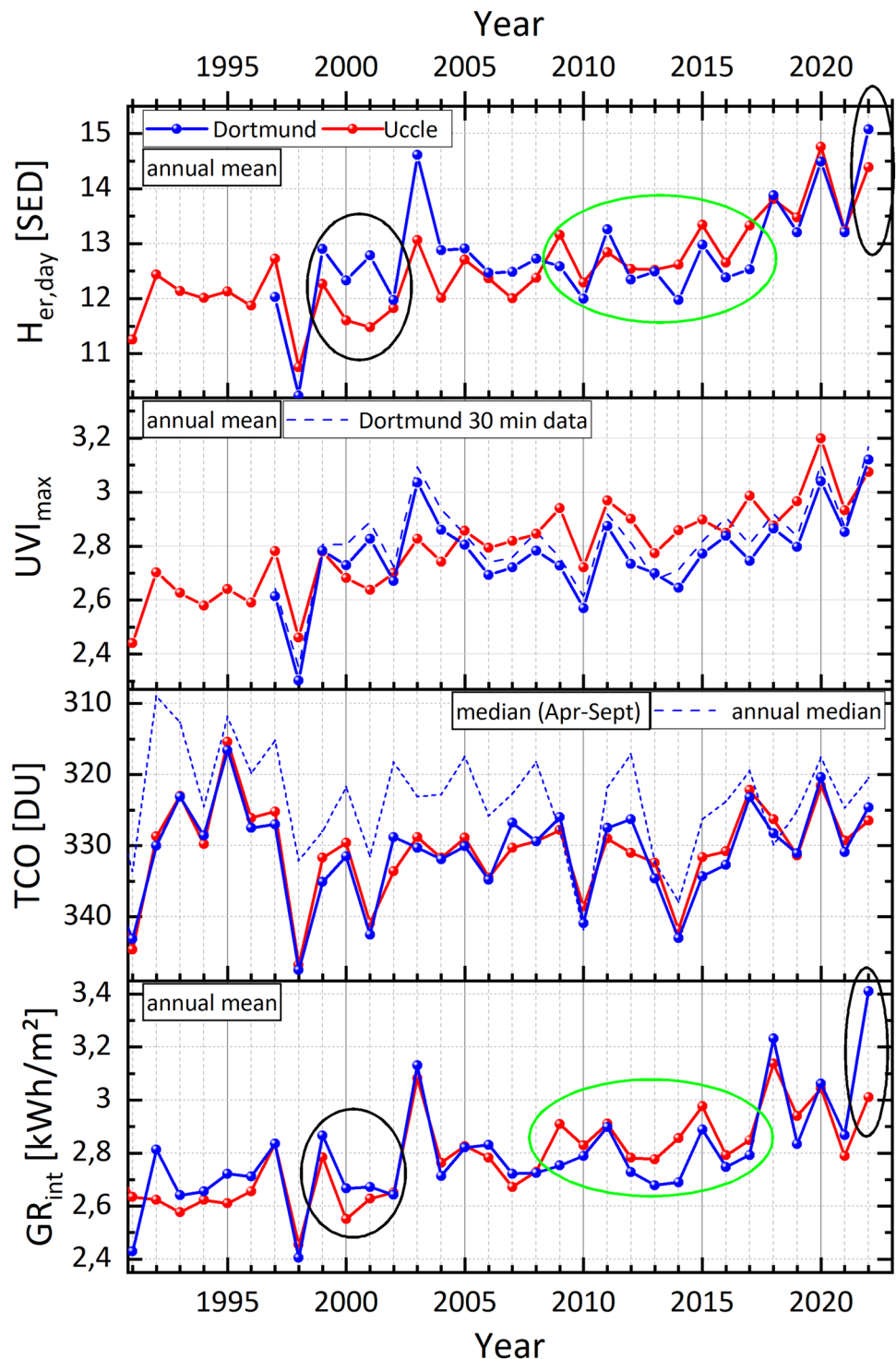


Fig. 16 Time series of annual mean of $H_{\text{er,day}}$ in Dortmund (red bars), annual mean of the daily global radiation (GR_{int} , black line) and annual median of summer ozone values (Apr–Sept, blue dashed line, inverted scale)

Fig. 17 Time series comparison of annual mean values from Dortmund (blue line) and Uccle (red lines) of $H_{\text{er,day}}$, daily UVI_{max} values with additional UVI_{max} values from Dortmund based on 30 min data (blue dashed line), annual median of TCO considering summer months (Apr–Sept, inverted scale) with additional annual TCO values (all month, blue dashed line) and annual mean of daily global radiation (GR_{int}); Black circles highlight periods where global radiation in Dortmund exceeded that in Uccle, while a green circle indicates an exemplary period during which global radiation in Uccle exceeded that in Dortmund. Annual mean values for both $H_{\text{er,day}}$ and UVI_{max} include imputation data for data gaps for both stations



2006, 2022) monthly means of $H_{\text{er,day}}$ and UVI_{max} in summer clearly stand out.

The estimated autoregressive order and the results of the Durbin–Watson test confirm that the monthly means of the UV data, even after considering the trend and seasonal pattern, are autocorrelated. In addition, the dispersion of the UV data in the summer months is higher than in the winter

months (Fig. 3, Fig. 20 in Appendix), demonstrating heteroskedastic behavior.

4.2.1 Trend results

Table 2 presents the trend results for UVI_{max} and $H_{\text{er,day}}$ in Dortmund and Uccle for the period 1997–2022 (26 years)

Fig. 18 Monthly means of $H_{\text{er,day}}$ from Dortmund based on partially missing daily values (grey circles) and based on partially imputed daily values (numbered circles; number = month); model fit (solid red line) and linear trend (dashed red line) for monthly means based on partially imputed daily values

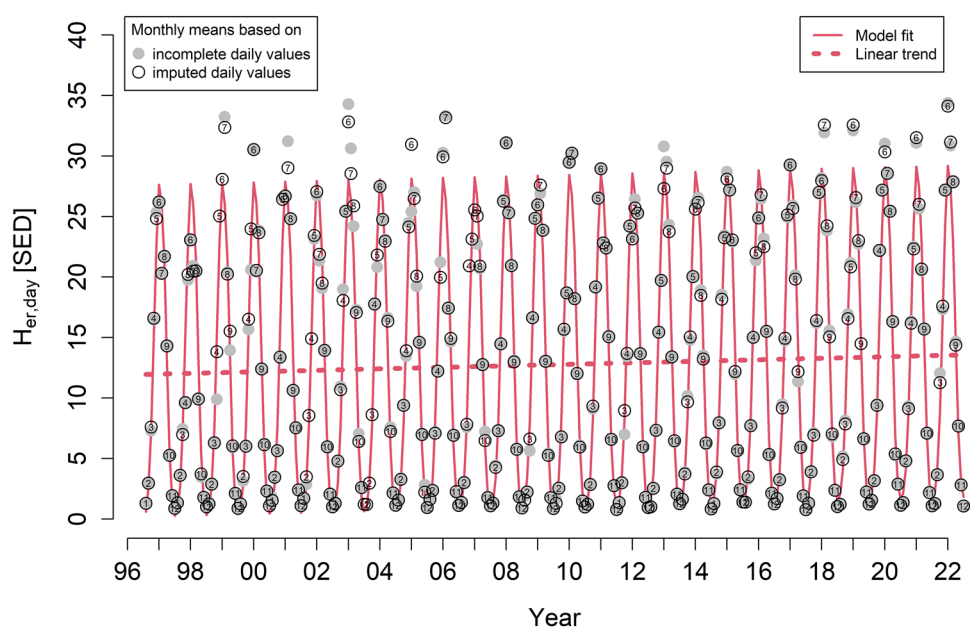


Fig. 19 Monthly means of UVI_{max} from Dortmund based on partially missing daily values (grey circles) and based on partially imputed daily values (numbered circles; number = month); model fit (solid red line) and linear trend (dashed red line) for monthly means based on partially imputed daily values

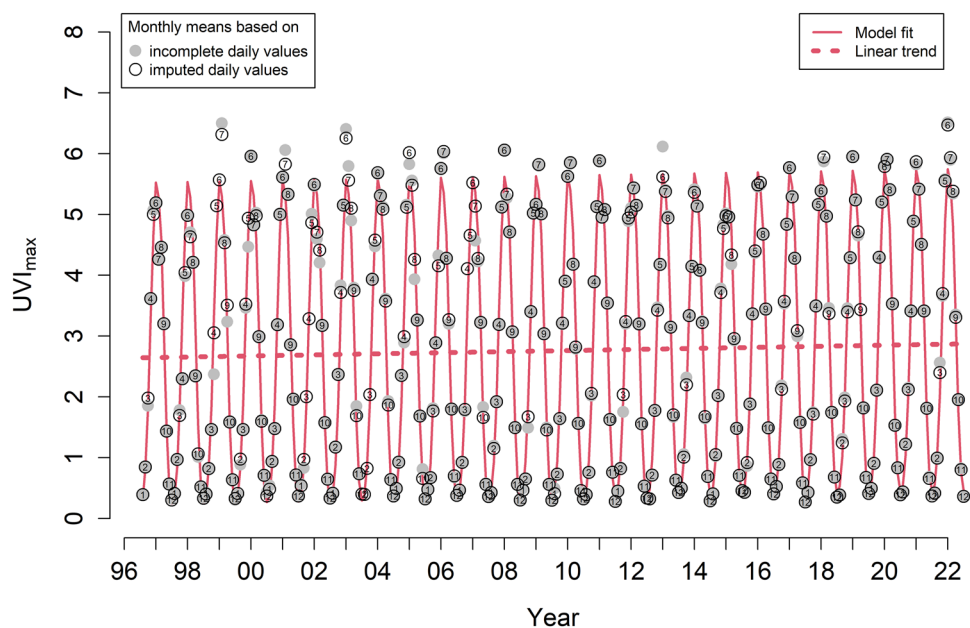


Table 2 Trend analysis results for UVI_{max} and $H_{\text{er,day}}$ (SED) for Dortmund and Uccle with imputation at data gaps and BFS trend model with standard error and 95% confidence interval (CI)

Location	Period	Dortmund		Uccle		Uccle	
		1997–2022 (26y)		1997–2022 (26y)		1991–2022 (32y)	
		UVI_{max}	$H_{\text{er,day}}$	UVI_{max}	$H_{\text{er,day}}$	UVI_{max}	$H_{\text{er,day}}$
Trend p. decade	abs	0.09	0.62 SED	0.16	0.94 SED	0.16	0.70 SED
Trend p. decade	[%]	3.2	4.9	5.8	7.5	5.6	5.6
Standard error	[%]	1.4	1.8	1.0	1.5	0.8	1.1
95% CI	[%]	0.4–6.0	1.4–8.4	3.7–7.8	4.6–10.4	4.0–7.2	3.5–7.8

Table 3 Trend analysis results for daily UVA and UVB radiant exposure for Dortmund with imputation at data gaps and BfS trend model with standard error and 95% confidence interval (CI)

Location		Dortmund		
Period		1997–2022 (26y)		
		UVA	UVB	UVA/UVB
Trend p. decade	abs	35.8 kJ/m ²	335 J/m ²	2.8
Trend p. decade	[%]	6.2	3.5	3.1
Standard error	[%]	1.5	1.7	1.0
95% CI	[%]	3.2–9.3	0.2–6.9	1.2–5.1

and additionally for Uccle from 1991 to 2022 (32 years). All trend analyses indicate statistically significant increases for UVI_{max} and $H_{er,day}$ as the corresponding 95% confidence intervals lie above zero. The observed trends are larger for Uccle than for Dortmund, and larger in $H_{er,day}$ than in UVI_{max} . However, considering the overlap of the confidence

intervals, the trends can be considered similar. Comparing the different periods (1997–2022 vs 1991–2022) for Uccle UVI_{max} the trend for UVI_{max} are nearly constant (5.8% vs 5.6%) while the trend for $H_{er,day}$ drops from 7.5% to 5.6%. Repeating the analysis with the data without imputation for missing data shows a slightly higher increase compared to the results with imputed data (see Table 5). Using the standard linear model on these data, results demonstrate the same percentage increase but the 95% CI is significantly smaller (see Table 6).

In addition to UVI_{max} and $H_{er,day}$, trends in the daily radiant exposure of UVA (315 nm to 400 nm) and UVB (290 nm to 315 nm), and the ratio of UVA to UVB radiant exposure were analyzed for Dortmund. Significant increases were found for both, with the UVA daily radiant exposure increasing almost twice as much as the UVB daily radiant exposure (Table 3). This is also reflected in the rise of the UVA-to-UVB ratio.

Table 4 Trend analysis results for monthly mean values of global radiation (GR_{int} , GR_{max}), daily sunshine duration (SunD) and total column ozone (TCO) for all month and for summer months (Apr–

Sept) for Dortmund with BfS trend model based on monthly mean with standard error and 95% confidence interval (CI)

Location		Dortmund					
Period		1997–2022 (26y)					
		GR_{max}	GR_{int}	SunD	SunD (Apr–Sept)	TCO	TCO (Apr–Sept)
Trend p. decade	abs	4.64 kWh/m ²	46.3 kWh/m ²	0.46 h	0.66 h	0.35 DU*	-2.98 DU
Trend p. decade	[%]	3.0	4.6	11.3	11.1	0.1*	-0.9
Standard error	[%]	0.9	1.5	2.3	2.5	0.5	0.4
95% CI	[%]	1.1–4.8	1.6–7.7	6.7–15.9	6.2–16.1	-0.8–1.1	-1.75–(-0.03)

Not significant trends are marked with *

Table 5 Trend analysis results for UVI_{max} and $H_{er,day}$ for Dortmund and Uccle without imputation at data gaps (Fig. 2a) and BfS trend model based on monthly mean with standard error and confidence interval (CI)

Location		Dortmund		Uccle		Uccle	
Period		1997–2022 (26y)		1997–2022 (26y)		1991–2022 (32y)	
		UVI_{max}	$H_{er,day}$	UVI_{max}	$H_{er,day}$	UVI_{max}	$H_{er,day}$
Trend p. decade	abs	0.10	0.71 SED	0.16	0.102 SED	0.16	0.99 SED
Trend p. decade	[%]	3.7	5.6	5.8	8.1	5.7	8.0
Standard error	[%]	1.5	1.8	1.0	1.6	0.8	1.3
95% CI	[%]	0.8–6.6	1.9–9.2	3.7–7.8	5.0–11.3	4.1–7.3	5.5–10.5

Table 6 Results of a standard trend analysis of UVI_{max} and $H_{er,day}$ for Dortmund and Uccle without imputation at data gaps (Fig. 2a) and standard linear trend model based on monthly anomalies with standard error and confidence interval (CI)

Location		Dortmund		Uccle		Uccle	
Period		1997–2022 (26y)		1997–2022 (26y)		1991–2022 (32y)	
		UVI_{max}	$H_{er,day}$	UVI_{max}	$H_{er,day}$	UVI_{max}	$H_{er,day}$
Trend p. decade	abs	0.10	0.71	0.16	1.02	0.16	0.99
Trend p. decade	%	3.7	5.6	5.8	8.1	5.7	8.0
Standard error	%	0.9	1.2	0.7	1.1	0.5	0.9
95% CI	%	1.9–5.4	3.2–7.9	4.4–7.1	5.9–10.3	4.7–6.7	6.1–9.7

Trends in the additional variables for the period 1997–2022 in Dortmund were analyzed (Table 4). A significant increase of a similar magnitude to that of UVI_{max} and $H_{er,day}$ was found for global radiation (GR_{int} and GR_{max}). For total column ozone data, no statistically significant change over the period was found when considering all ozone data. A statistically significant decrease in total column ozone is observed during the summer months. The most substantial increase is observed in the daily sunshine duration (SunD), with a 11.3% per decade.

4.2.2 Impact of data gap handling and trend model

The influence of improved data gap handling on the outcome of trend analysis can be demonstrated by repeating the calculations using the dataset with gaps (e.g. Figure 2a). Table 5 presents the results of these calculations based on UVI_{max} and $H_{er,day}$ data from Dortmund and Uccle. Both for Dortmund and Uccle, the outcome indicates a slightly higher increase compared to the results presented in Table 2.

The influence of the improved BfS trend model on the outcome of trend analysis can be demonstrated by repeating the calculations using the dataset with gaps and a standard linear model applied to the monthly anomalies. The results (Table 6) demonstrate the same percentage increase (compared to Table 5); however, the 95% confidence interval is significantly smaller.

5 Discussion

This study is the first comprehensive analysis of data from a station of the German Solar UV Monitoring Network. It shows distinct changes in ground-level solar UVR and its influencing factors, in particular global radiation and total column ozone. The trend analysis shows a clear increase in UV exposure across nearly three decades.

5.1 UV exposure in Dortmund

The dispersion of daily UV data (e.g. Figure 3) primarily arises from the cloud cover situation, with greater solar elevation leading to increased absolute dispersion. The differing fluctuations contribute to the heteroscedasticity of the measurement data. When comparing the monthly mean values of unweighted UVA and UVB (Figs. 5, 6) to literature data, they are in the expected order of magnitude. For example, they appear slightly lower than values observed in

Vienna [29]. The main contributing factors to these differences are primarily latitude and varying typical cloud cover situations.

Looking at the annual frequency of daily UVI_{max} values shows that approximately half of the days in Dortmund have a UVI_{max} value ≥ 3 (Fig. 8). According to WHO recommendations, sun protection is necessary for these days [11]. Days with UVI_{max} values of 3, 4, 5, and 6 occur with roughly equal frequency. The number of days with a UVI_{max} of 7 is significantly lower, and days with a UVI_{max} of 8 are rare and restricted to specific combinations of influencing factors (total column ozone, solar elevation, cloud cover). UVI_{max} values ≥ 3 occur mainly (94%) between April and September. On average, 83% of the annual cumulative $H_{er,day}$ occurs from April to September.

Approximately one-third of the days per year falls into the range of 0–5 daily SED $H_{er,day}$ (Fig. 8, Fig. 9). This quantity corresponds to the number of days with UVI_{max} value 0 and UVI_{max} value 1, as expected from the scatterplot of $H_{er,day}$ and UVI_{max} (Fig. 14). This scatterplot also demonstrates that days with low or no cloud cover (red dots), are at the upper boundary of the pattern and show an approximate linear relationship between $H_{er,day}$ and UVI_{max} . The slope is nearly identical to the evaluation of data from Spain [41] and higher than in the evaluation of data from New Zealand [81]. This discrepancy is attributed to differences in daylight duration, which result in higher $H_{er,day}$ values for the same UVI_{max} in Spain and Germany. Consistent with findings in other studies, it is observed that on days with $UVI_{max} < 3$ (2.5), the daily accumulation of $H_{er,day}$ values can exceed 10 SED. A closer examination reveals that even at noon on these days, a value of 2.5 SED per hour is not reached. The threshold of 2.5 SED per hour is considered as the limit for visible skin damage for fair skin, as proposed by Fitzpatrick [81, 82]. Cloudless days with $UVI_{max} < 3$ (2.5) and $H_{er,day} > 10$ SED usually occur in March and September/October in Dortmund/Uccle. Our findings thus affirm the practicability of the WHO recommendations for protection for $UVI_{max} \geq 3$ [11].

Considering the annual courses of potential highest UVI_{max} and $H_{er,day}$ values, a shift to the summer solstice towards later months can be observed. Like at the UVA/UVB ratio course (Fig. 7), this shift can be attributed to the annual course of ozone values, which are independent of solar elevation. In the months around and after the summer solstice, particular attention must be paid to the WHO sun protection measures [11], as the highest UVI_{max} and $H_{er,day}$ in the year could be expected here.

In addition to annual course of normal ozone, low ozone events can significantly influence ground-level UVR [59–62], especially in absence of clouds. For the observed low ozone events occurring in Dortmund (Fig. 15), the UVI_{max} values increase with increasing negative deviation

from the ozone mean value (for cloudless conditions and similar solar maximum elevation). In winter, this is compounded by the fact that changes in TCO have a stronger influence on UVI_{max} at ground level due to the longer path of UVR through the atmosphere. Hence, the high percentage UVI_{max} enhancement of over 50%, as seen in February 2023, is not expected during the summer months. It has to be remarked, that for the reported cases the aerosol situation was not considered. Aerosol effects can counteract the UVI_{max} increase at low ozone events [62, 74]. However, since lower ozone in Dortmund usually leads to a greater UVI_{max} increase, (for cloudless conditions and similar solar maximum elevation) it can be assumed that aerosol effect on the low ozone events is at least a small or even negligible in Dortmund.

From a radiation protection perspective, it is important that the observed superelevations often occur following a period of inclement weather with low UVI_{max} values. UVI_{max} exposure then rises sharply, and a low ozone event significantly amplifies this increase. In spring, it is further compounded by the fact that the solar elevation angle exhibits the highest day-to-day increase over the course of the year. This contributes to a further elevation in the typically abrupt rise in UV levels. Some studies show that under expected climatic changes, low ozone events might become more frequent in the future during Northern Hemisphere spring [83, 84].

5.2 Trend in UV exposure

For monthly mean values of UVI_{max} and $H_{er,day}$, the results indicate a clear increase that is statically significant at both locations (Table 2). For Dortmund, the magnitude of the trend in UVI_{max} and $H_{er,day}$, as well as the difference between both values, aligns well with the change in daily maximum and daily integral of global radiation (Table 4). An increase of global radiation can be primary traced back to a decrease/change of aerosols, clouds and their interactions [85–88]. In the case of aerosols, one distinguishes between direct aerosol effects (e.g. changing AOD), semi-direct effects and indirect effects [88]. The semi-direct and indirect effects are linked with aerosol–cloud interaction. In case of clouds, various studies indicate that changes in cloud cover are also influenced by greenhouse gases (GHGs) and climate change [89, 90]. The specific proportion of cloud and aerosol cause of the global radiation increase in the examined period (1996–2022) at the locations Dortmund and Uccle cannot be estimated within the scope of this study. However, the strong increase in monthly mean of local sunshine duration compared to the global radiation trend (Table 4) suggests a strong effect of decreasing cloud cover in Dortmund, as sunshine duration almost exclusively depends on cloud cover and negligible on AOD. In addition, the correlation

coefficient of sunshine duration with $H_{er,day}$ is higher than of sunshine duration with UVI_{max} (Fig. 12). A change in sunshine duration thus has a stronger effect on $H_{er,day}$ than on UVI_{max} which is consistent with the different $H_{er,day}$ and UVI_{max} trends. Regarding the UV influencing parameter ozone, the results show a slight but significant decrease in the summer months (Table 4). Due to the much more significant change in global radiation, it can be concluded that the changes in monthly mean UVI_{max} and $H_{er,day}$ values are primarily driven by a influencing factor also affecting the global radiation. Considering the trend in SunD, this indicates a change in cloud cover as a main driver. The decreasing summer ozone certainly influences UVI_{max} and $H_{er,day}$ changes; however, it is just a minor driver.

When comparing the locations, it becomes evident that in Uccle, the trend in monthly mean $H_{er,day}$ and UVI_{max} values is higher than in Dortmund. This difference was to be expected from the comparison of the annual means (Fig. 17), as in Dortmund, the annual UV values were mostly higher than those in Uccle from 1999 to around 2004 and lower from 2013 to 2017. In the years 2018–2021, the annual mean values of $H_{er,day}$ for both stations were comparable. However, the UVR levels at both stations for the entire period are similar (Fig. 3). Higher annual $H_{er,day}$ values in Uccle are associated with higher annual global radiation in Uccle and vice versa (Fig. 17). Regarding the annual means of UVI_{max} , Uccle tends to have higher values than Dortmund. This difference partly arises from distinct measurement intervals, which result in slightly higher UVI_{max} values in Uccle. However, this effect alone does not account for the observed differences, as evidenced by a UVI_{max} calculation for Dortmund with a 30-min interval (dashed blue UVI_{max} line in Fig. 17). Additional differences may be attributed to localized cloud cover effects at solar noon. McKenzie et al. [91] see differences in the use of single maximum values instead of mean maximum values for the UVI_{max} determination and attribute this to reduced cloud effects in the case of single maximum values.

Based on the trend analysis of the longer period 1991–2022 (32 years) for Uccle, it can be observed that the trends for the monthly mean UVI_{max} values remain relatively stable compared to the 1997–2022 analysis but with a smaller 95% CI for the longer period (Table 2). The change in monthly mean $H_{er,day}$ values calculated for the longer period (1991–2022) is identical to the change in UVI_{max} values and lower than the $H_{er,day}$ change calculated for the period 1997–2022. It has to be remarked that different $H_{er,day}$ trends of Uccle for the two periods (1991–2022 vs. 1997–2022) only become apparent when using data set with imputed data (Table 5).

In the results of the trend analysis of unweighted UVA and UVB radiation (monthly mean of daily radiant exposure) in Dortmund, significant increases and an almost twice as

large trend in UVA radiation as compared to UVB radiation are observed (Table 3). Consequently, the UVA-to-UVB ratio also increases significantly. As expected, the trend per decade in $H_{\text{er,day}}$ (Table 2) falls between the changes in UVA and UVB. The ozone trend analysis excluded the parameter ozone as a main driver for the trend of the UVA/UVB ratio. Decreasing AODs [86] are also unlikely to be the cause of our UVA/UVB ratio trend result. Although aerosol radiative impact depends on the amount, size, and type of the atmospheric particles, what can change the ratio of UVA–UVB through various effects, the scattering of aerosols is generally stronger at shorter wavelengths. This effect remains even with decreasing AOD even if the absolute scattering intensity decreases, which therefore still leads to a relatively stronger reduction in UVB than UVA scattering at a given AOD. Because of the overall stronger scattering at shorter wavelengths, a decreasing AOD trend would therefore result in a relatively greater increase in UVB over UVA at ground level. For example, Fountoulakis et al. [38] analyzed a long-term ground-level UV dataset from Thessaloniki and demonstrated a disproportionately high increase in UVB radiation with decreasing AOD levels. At least a portion of the different increase in UVA and UVB can be attributed to the strong increase in sunshine duration because the investigated correlations between sunshine duration and UVA as well as sunshine duration and UVB indicate a slightly higher correlation coefficient between sunshine duration and UVA throughout the year (Fig. 13). Additionally, the greater increase in UVA radiation compared to global radiation (GR_{int}) is consistent with decreasing cloud cover [92]. The extent to which indirect aerosol effects play a role, if any, and how significant they are, is an important open question that needs to be examined in other studies.

When looking at the observed long-term changes in ground-level UVR across all of Europe, a highly heterogeneous pattern emerges. Influencing factors (clouds, total ozone column, AOD) have different effects at different locations and with respect to the large differences there are only very few high-quality long-term measurement series in Europe [32–43]. With the detailed comparison of the results from Dortmund with those from Uccle, statements on the significant increase of ground-level UVR on a similar latitude in central Europe are confirmed. With Uccle UV data of the period from 1996 to 2017, Fountoulakis et al. [36] found an increase of 5% per decade for UVB (307.5 nm) which is in similar magnitude to our findings. An increasing trend in Uccle was also found by De Bock et al. analyzing the $H_{\text{er,day}}$ monthly mean values of 1991–2013 (+7% p. decade) [37]. Decreasing attenuation by AOD and cloud cover have so far been identified as main contributors in Uccle [33, 37]. Further analyses across Europe range from no significant long-term trend in Finland [32, 33] to a decreasing trend of 7–8% per decade in the UK [34–36], to positive trend of up to 5%

per decade in Rome in some months [39], and a strongly positive trend in Thessaloniki of 8% per decade [36]. The trend in Thessaloniki was mainly driven by changes in aerosols [38], indications of significant changes in cloud cover and ozone were not found [33].

In contrast to annual ozone values, the annual summer ozone values show a decreasing trend (Fig. 16 and Table 4). A 1% change in total ozone column can lead to a significant change of spectral irradiance in the UVB range (about 3% for 300 nm, 7% for 295 nm) [30] and an increase of around 1.2% in erythral irradiance [93, 94]. For every 1% total ozone column decrease, the incidence of melanoma is projected to rise between 1 and 2%, the incidence of squamous cell carcinoma between 3% and 4.6% and basal cell carcinoma between 1.7% and 2.7% [95].

Our ozone trend results (Table 4) are consistent with other studies: In northern mid-latitudes (35–60° N), no trend in annual mean total ozone column is observed [56]. A study by Malinović-Milićević et al. [49] analyzed ozone trends in Novi Sad, Serbia separated by seasons. They observed statistically significant negative trend of annual ozone values for summer (−0.8% p. decade), spring (−0.5% p. decade), and autumn (−1.0% p. decade) over the period 1997–2018. Similarly, in a study by Fountoulakis et al. [39], significant negative trends in total ozone were reported for Rome in April and September for the period 1996–2020. They noted that these trends coincided with positive trends in Geopotential Height (GPH) over extensive regions, indicating an upward shift of the tropopause to higher altitudes. This is congruent with anticorrelation of the tropopause height and the total ozone column [96]. A rising altitude of the tropopause can be attributed to climate change associated with the warming of the troposphere [97, 98]. Steinbrecht et al. discussed in detail the possible causes of the absence or delay of an expected increasing trend in total column ozone, including the influence of climate change [99], which was also addressed by Langematz [100].

The lack of annual ozone trend and negative summer ozone trend could be also explained by a temporal shift of the annual course of ozone towards earlier months (see Fig. 21 in appendix). At such a shift, lower ozone values occur between April and September while the annual average remains unchanged. Such a temporal shift would result in ozone values tending to decrease during the period of highest solar elevation in the year, potentially leading to conditions for higher UVI_{max} and $H_{\text{er,day}}$ values.

5.3 Impact of data gap handling and trend model

In this study, the issue of data gap handling was addressed using a developed and validated imputation method [73]. The applied imputation method prevents a significant bias due to data gaps. To demonstrate the impact of the

imputation method on trend results, additional trend analyses were conducted using a dataset containing data gaps (for Dortmund: Fig. 2). The results indicate that in most cases, there is an increased trend per decade (Table 5) compared to the results obtained with imputation (Table 2). Standard errors remain constant or similar. The increasing trends are likely attributed to the temporal appearance of the data gaps. Changes in the trends are traceable to both to the missing months in the data set of Dortmund and to monthly means with missing daily values. In a different temporal gap appearance, a decrease in trend per decade could also be possible.

In our trend analysis, we improved the standard linear model to account for autocorrelation and heteroscedasticity present in our datasets. An additional advantage of our approach is the direct use of monthly mean values, bypassing the need for monthly anomalies. For comparison, the $H_{\text{er,day}}$ and UVI_{max} data (with gaps) were analyzed using a standard linear model based on monthly anomalies. The results Table 6 as reveal nearly identical changes/trends per decade compared to the advanced model without imputation at data gaps (Table 5), as expected due to similar slope calculations. However, a distinction arises in the larger standard error and significantly expanded 95% confidence interval when using the advanced BfS model. The non-consideration of existing autocorrelation and heteroscedasticity would lead to an underestimation of the standard errors of the trend estimates. In general, this becomes crucial when assessing the significance of trend results with a low change per decade.

6 Conclusion

In this study, a 26-year series of spectrally resolved data on the solar UVR from Dortmund, Germany, was evaluated and compared to Uccle, Belgium. Increasing trends were detected. Increasing solar UVR can lead to higher human UV exposure and can adversely affect the environment.

- The trend analysis shows a significant increase in $H_{\text{er,day}}$ and UVI_{max} in Dortmund, Germany, for the period of 1997–2022, with even higher values in Uccle, Belgium. In Dortmund, the rise in both $H_{\text{er,day}}$ and UVI_{max} correlates well with the trend in global radiation. This together with a strong trend in sunshine duration indicates a change in cloud cover as the primary driver. In terms of radiation protection, an impact of the $H_{\text{er,day}}$ and UVI_{max} trends on public health has to be expected thus cannot be numerated.
- A statistically significant decrease in summer ozone (Apr–Sept) levels contributes to higher UVB, UVI_{max} and $H_{\text{er,day}}$ values. If this trend continues, we have to expect earlier and longer periods of high solar UVR. Low

ozone events result in a superelevation of UVI_{max} values in the magnitude of 10–20% in the summer half-year; an extremely low ozone event in Feb. 2023 leads to an increase in UVI_{max} of over 50%.

The findings of this study underscore the importance of developing additional measures to counter the rising UV exposure in Europe. To achieve this objective, further research is required into the influence of climate change and the interaction of the main influencing factors. The observed trends must be factored into future adaptive and preventive strategies against the current climate change to minimize health-related consequences effectively.

Appendix

Trend analysis

Details of trend model

The model (Eq. (1)) in Sect. 3.2 can be written in matrix notation as:

$$y = X\beta + \epsilon \quad (2)$$

with the column vector of response variable values $y = (y_1, \dots, y_n)^T$ from the first to the last month n , the error vector $\epsilon = (\epsilon_1, \dots, \epsilon_n)^T$, the coefficient vector $\beta = (\beta_0, \dots, \beta_{12})^T$ and the $n \times 13$ design matrix

$$X = \begin{pmatrix} 1 & t_1 & \dots \\ \vdots & \vdots & \dots \\ 1 & t_n & \dots \end{pmatrix} \quad (3)$$

Here, the superscript index T denotes the transposed vector or matrix. The values $t_1, \dots, t_n$ counts the months from 1 (Jan 1990) to 396 (Dec 2022). This means, for instance, that $t_1 = 85$ and $t_n = 396$ for analyses in the period from January 1997 to December 2022. With the assumption of normally distributed errors, the maximum likelihood (ML) estimate for the coefficient vector β is given by

$$\hat{\beta} = (X^T X)^{-1} X^T y \quad (4)$$

whereas in the standard linear model uncorrelated error variables with the same variance σ^2 for all months are assumed, we used a linear model that provides a heteroskedasticity and autocorrelation consistent estimation of the general covariance matrix Ω of $\epsilon \sim N(0, \Omega)$, which have been suggested in the econometric literature (e.g. [101, 102]) and implemented in the package ‘sandwich’ for the software R by Zeileis [80]. Thus, for the covariance matrix of $\hat{\beta}$ one gets

$$\text{Cov}(\hat{\beta}) = (X^T X)^{-1} X^T \hat{\Omega} X (X^T X)^{-1} \quad (5)$$

with

$$X^T \hat{\Omega} X = E \left(v(\hat{\beta}) (v(\hat{\beta}))^T \right) \quad (6)$$

where $v(\beta) = X^T(y - X\beta)$. Assuming an autoregressive structure with a weight vector $w = (w_0, w_1, \dots, w_{n-1})$, a heteroscedasticity and autocorrelation consistent estimation of the covariance matrix of the coefficient estimates in a linear regression model is given by Zeileis [80]:

$$X^T \hat{\Omega} X = \sum_{s,t} w_{|s-t|} v_s(\hat{\beta}) (v_t(\hat{\beta}))^T \quad (7)$$

This approximation is done after the coefficient vector β has been estimated. We determine the weights for specific lags l as proposed by Andrews [102]:

$$w_l = \frac{3}{z^2} \left(\frac{\sin z}{z} - \cos z \right) \quad (8)$$

with $z = 6\pi/5 \cdot l/B$. The bandwidth B is determined adaptively, analogous to Andrews [102]. Unlike fixed or pre-determined bandwidths, the adaptive method adjusts the bandwidth dynamically based on the characteristics of the data itself.

The autocorrelation structure of the response values was examined by analysis of the residuals. For this purpose, the autoregressive order of the residuals was estimated and a Durbin–Watson test [103] was applied. Regarding homogeneity of variance, it was investigated whether the residual

plot (residuals vs. fitted values) exhibits "chaotic" behavior. Additionally, it was tested whether the residual plot (residuals vs. time, grouped by months) displays the same dispersion in all months.

Residuals

See Fig. 20

Measurement uncertainties in the trend analysis

The correlated contributions to the measurement uncertainty, which remain approximately constant for a specific month over the years, and include potential systematic errors e.g. due to diffuser temperature or angular response, are incorporated into the month term of the trend model for both stations. The remaining contributions to measurement uncertainty are mainly related to the radiometric uncertainty set during calibrations. As mentioned in Sect. 2.1, this radiometric uncertainty (<2% expanded) for the Dortmund station is randomly set during calibrations conducted at least once a year. We therefore assume this remaining (radiometric) uncertainty has negligible impact on the results of the trend analysis. We apply the same assumption to the trend analysis of the Uccle data. Although the remaining contributions to measurement uncertainty are expected to be somewhat higher here than in Dortmund, we still consider the assumption justified due to the monthly check and the presence of a mostly concurrently operating second measurement system explained in Sect. 2.2.

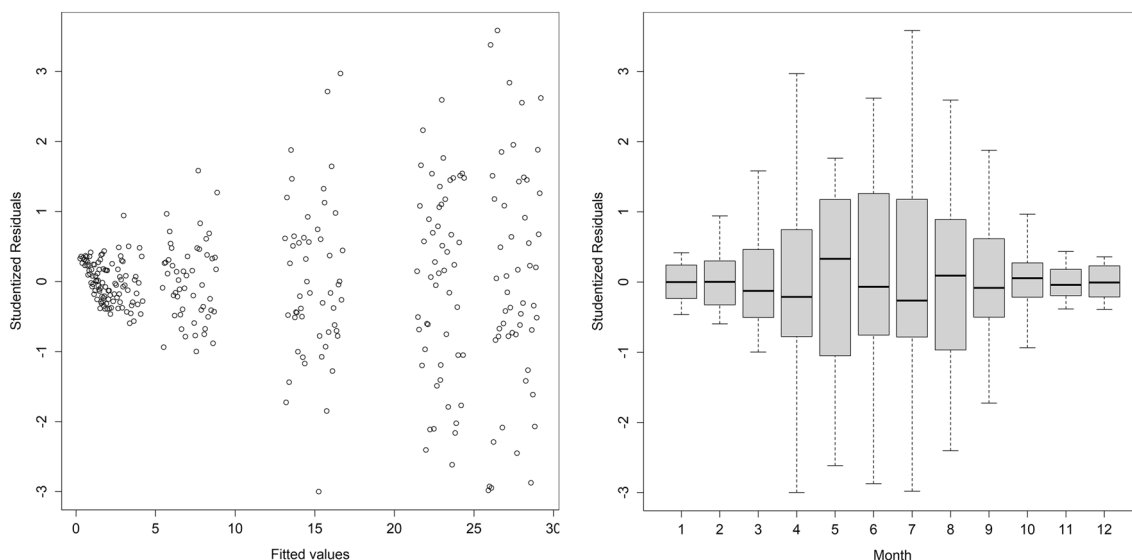


Fig. 20 Studentized residuals depending on the fitted values (left) and months (right) for $H_{\text{er,day}}$ [SED] values from Dortmund calculated with imputation at data gaps (Fig. 2) and BfS trend model (chapter 3.2)

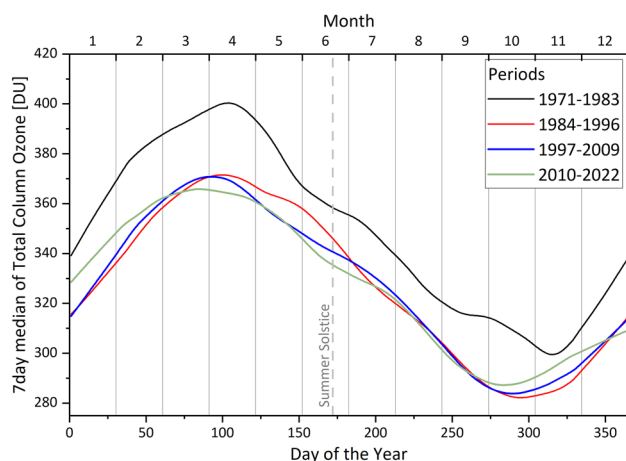


Fig. 21 Annual courses of 7-day-median ozone in Dortmund for different 13-year periods: 1971–1983 (black), 1984–1996 (red), 1997–2009 (blue) and 2010–2022 (green) smoothed by LOESS filter

Development of annual course of ozone

To investigate a possible change of the annual course of ozone values, the annual median course of ozone was separately calculated for the first (1997–2009) and second half (2010–2022) of the period. Additional TEMIS data from 1984 to 1996 and 1971 to 1983 were used to calculate two further 13-year periods (Fig. 21). The annual courses are smoothed by a LOESS filter for a better visualization of changes. It is clear to see how the Montreal Protocol (1987/1989) halted the significant decline of ozone throughout the entire year. Subsequently (red, blue and green line), only a minor decrease in ozone is observable in spring and summer, accompanied by a slight increase in autumn and winter. This change is consistent with a temporal shift of the annual course towards earlier times. Such a temporal shift is particularly obvious between the last two periods 1997–2009 (blue line) and 2010–2022 (green line). Further research is needed, on how the parameters influencing total column ozone affect the annual course of ozone.

It is worth noting that when comparing ozone levels across different time periods, the solar cycle (SC) can also be a contributing factor. To exclude bias resulting from the choice of start and end points, time periods of 1976–1985 (SC21), 1986–1996 (SC22), 1997–2008 (SC23), and 2009–2020 (SC24) were compared, roughly coinciding with completed solar cycles, instead of using fixed 13-year periods. A difference can only be seen in the minor fluctuations along the curves. The tendency of a temporal shift can be shown in the same manner.

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Data availability Researchers interested in accessing our data can contact us via the corresponding author. We will handle inquiries on a case-by-case basis, as different requests may require different options and responsibilities. Please rest assured that the data is available for scientific purposes.

Declarations

Conflict of interest The authors declare that they have no conflict of interests.

Sustainability This study focuses on ground-level solar UV radiation and identifies both changes and their potential causes. The research represents a significant contribution to the United Nations' 2030 Agenda Sustainable Development Goals (SDGs). It specifically addresses the SDGs 3 (Good health and well-being), 11 (Sustainable cities and communities), 13 (Climate action) and 15 (Life on land). Barnes et al. [104] provide a comprehensive overview of direct and indirect connections between the topic and further SDGs.

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